FlexBridge 1.3

Baza, imkoniyatlar va API atlasi

94 jadval | 923 maydon | 205 FK

14 rol | 40 imkoniyat | 124 asosiy va 8 yordamchi API

Barcha maydon va parentlar chizmalarda va lug‘atda.

To‘liq umumiy ER va interaktiv havolalar HTML/SVGda.

Bu mantiqiy loyiha; dastur va real migratsiya hali tekshirilmagan.

30 oktabr 2026

Mundarija va o‘qish tartibi

Domainlar bo‘yicha to‘liq ER chizmalari **3**

User -> imkoniyat -> baza xaritasi **15**

To‘liq ma’lumotlar bazasi lug‘ati **19**

API amallari kontrakt jadvali **74**

DB-xxx: jadval, CAP-xxx: imkoniyat, API-xxx: method/path.

Har jadvalda barcha maydon, type, NULL, PK va aniq parent FK bor.

Scope/immutable/capacity qoidalari asosiy topshiriqdagi INV-01..16da.

Chizmalarni kattalashtirish uchun offline HTMLni oching.

IAM | Hisob, kirish va ruxsatlar

Barcha maydonlar. FK yonidagi DB indeksi parent.idni bildiradi. Tashqi domainlar umumiy SVG/Mermaidda bog‘langan.

PK = primary key; ? = NULL; FK child -> parent. Parentda 0..N child, non-null FKda child uchun aynan1 parent.

DB-001
users

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
display_name	text
status	text
password_hash ?	text
locale	text
timezone	text
adult_declared_at ?	timestamptz
last_login_at ?	timestamptz

DB-002
user_contacts

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK user_id	DB-001
channel	text
normalized_value	text
verified_at ?	timestamptz
revoked_at ?	timestamptz
is_primary	boolean

DB-003
auth_challenges

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK user_id ?	DB-001
FK contact_id ?	DB-002
destination_digest	text
channel	text
purpose	text
code_hmac	text
expires_at	timestamptz
attempts	integer
consumed_at ?	timestamptz
superseded_at ?	timestamptz

DB-004
sessions

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK user_id	DB-001
token_digest	text
device_key_digest ?	text
assurance	text
expires_at	timestamptz
revoked_at ?	timestamptz
last_seen_at	timestamptz

DB-005
mfa_credentials

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK user_id	DB-001
kind	text
credential_key	text
secret_encrypted ?	text
public_key ?	text
sign_count	integer
verified_at ?	timestamptz
revoked_at ?	timestamptz

DB-006
recovery_requests

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK user_id ?	DB-001
kind	text
status	text
requester_contact_digest	text
FK reviewer_id ?	DB-001
FK evidence_file_id ?	DB-018
resolved_at ?	timestamptz

DB-007
permission_catalog

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
code	text
description	text

DB-008
role_catalog

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
code	text
scope_kind	text
description	text

DB-009
role_permissions

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK role_id	DB-008
FK permission_id	DB-007

DB-010
platform_role_grants

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK user_id	DB-001
FK role_id	DB-008
FK granted_by_id	DB-001
expires_at	timestamptz
revoked_at ?	timestamptz

DB-011
consent_events

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK user_id	DB-001
purpose	text
policy_version	text
action	text
legal_basis	text
occurred_at	timestamptz

CAN | Nomzod, malaka va vaqt

Barcha maydonlar. FK yonidagi DB indeksi parent.idni bildiradi. Tashqi domainlar umumiy SVG/Mermaidda bog‘langan.

PK = primary key; ? = NULL; FK child -> parent. Parentda 0..N child, non-null FKda child uchun aynan1 parent.

DB-012
candidate_profiles

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK user_id	DB-001
headline	text
region	text
bio ?	text
experience_months	integer
visibility	text
FK cv_file_id ?	DB-018

DB-013
skills

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
code	text
name_uz	text
name_ru	text
aliases	jsonb
active	boolean

DB-014
candidate_skills

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK candidate_id	DB-012
FK skill_id	DB-013
level	text
FK verification_case_id ?	DB-019

DB-015
enrollments

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK candidate_id	DB-012
FK institution_id	DB-028
program	text
student_ref_digest ?	text
start_date	date
end_date ?	date
status	text
FK verification_case_id ?	DB-019

DB-016
availability_rules

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK candidate_id	DB-012
weekday	integer
start_minute	integer
end_minute	integer
valid_from	date
valid_to ?	date
timezone	text

DB-017
availability_exceptions

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK candidate_id	DB-012
local_date	date
kind	text
start_minute	integer
end_minute	integer

DB-018
files

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK owner_user_id	DB-001
storage_key	text
filename	text
mime_type	text
byte_size	integer
sha256	text
scan_status	text
retention_class	text
deleted_at ?	timestamptz

DB-019
verification_cases

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK requester_id	DB-001
FK candidate_id ?	DB-012
FK organization_id ?	DB-021
FK institution_id ?	DB-028
FK enrollment_id ?	DB-015
FK appointment_id ?	DB-032
kind	text
status	text
current_revision	integer
FK assigned_reviewer_id ?	DB-001
due_at	timestamptz
decision_reason ?	text
decided_at ?	timestamptz

DB-020
verification_evidence

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK case_id	DB-019
revision	integer
FK file_id ?	DB-018
note ?	text

ORG | Tashkilot va xodimlar

Barcha maydonlar. FK yonidagi DB indeksi parent.idni bildiradi. Tashqi domainlar umumiy SVG/Mermaidda bog‘langan.

PK = primary key; ? = NULL; FK child -> parent. Parentda 0..N child, non-null FKda child uchun aynan1 parent.

DB-021
organizations

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
tax_id	text
legal_name	text
public_name	text
region	text
description ?	text
website ?	text
status	text

DB-022
organization_capabilities

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK organization_id	DB-021
capability	text
status	text
FK approved_case_id ?	DB-019

DB-023
org_memberships

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK organization_id	DB-021
FK user_id	DB-001
status	text
accepted_at ?	timestamptz
revoked_at ?	timestamptz

DB-024
org_role_grants

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK membership_id	DB-023
FK role_id	DB-008
FK granted_by_id	DB-001
revoked_at ?	timestamptz

DB-025
org_invitations

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK organization_id	DB-021
FK inviter_membership_id	DB-023
contact_digest	text
token_digest	text
expires_at	timestamptz
FK accepted_by_id ?	DB-001
accepted_at ?	timestamptz
revoked_at ?	timestamptz

DB-026
org_invitation_roles

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK invitation_id	DB-025
FK role_id	DB-008

DB-027
org_ownership_events

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK organization_id	DB-021
FK new_membership_id	DB-023
FK previous_membership_id ?	DB-023
FK verification_case_id	DB-019
FK approved_by_id	DB-001
effective_at	timestamptz

EDU | Muassasa va rasmiy biriktirish

Barcha maydonlar. FK yonidagi DB indeksi parent.idni bildiradi. Tashqi domainlar umumiy SVG/Mermaidda bog‘langan.

PK = primary key; ? = NULL; FK child -> parent. Parentda 0..N child, non-null FKda child uchun aynan1 parent.

DB-028
institutions

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
registry_code	text
name	text
region	text
status	text

DB-029
institution_agreements

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK institution_id	DB-028
version	integer
status	text
valid_from	date
valid_to ?	date
purposes	jsonb
FK signed_file_id ?	DB-018

DB-030
institution_memberships

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK institution_id	DB-028
FK user_id	DB-001
status	text
accepted_at ?	timestamptz
revoked_at ?	timestamptz

DB-031
institution_role_grants

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK membership_id	DB-030
FK role_id	DB-008
FK appointment_id	DB-032
revoked_at ?	timestamptz

DB-032
institution_appointments

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK institution_id	DB-028
FK user_id	DB-001
FK document_file_id	DB-018
status	text
FK verification_case_id ?	DB-019
valid_from	date
valid_to ?	date
accepted_at ?	timestamptz

DB-033
institution_invitations

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK appointment_id	DB-032
contact_digest	text
token_digest	text
expires_at	timestamptz
FK accepted_by_id ?	DB-001
accepted_at ?	timestamptz
revoked_at ?	timestamptz

DB-034
student_supervisions

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK enrollment_id	DB-015
FK coordinator_membership_id	DB-030
FK agreement_id	DB-029
purpose	text
status	text
legal_basis	text
accepted_at ?	timestamptz
revoked_at ?	timestamptz

JOB | E’lon va qidiruv

Barcha maydonlar. FK yonidagi DB indeksi parent.idni bildiradi. Tashqi domainlar umumiy SVG/Mermaidda bog‘langan.

PK = primary key; ? = NULL; FK child -> parent. Parentda 0..N child, non-null FKda child uchun aynan1 parent.

DB-035
opportunities

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK organization_id	DB-021
kind	text
status	text
FK current_revision_id ?	DB-036
capacity	integer
occupied_count	integer
published_at ?	timestamptz
expires_at ?	timestamptz
closed_at ?	timestamptz

DB-036
opportunity_revisions

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK opportunity_id	DB-035
revision	integer
FK author_id	DB-001
title	text
description	text
region	text
work_mode	text
eligibility	text
program_required ?	text
salary_min ?	numeric(18,2)
salary_max ?	numeric(18,2)
currency	text
salary_period ?	text
salary_basis ?	text
is_paid	boolean
start_date	date
end_date ?	date
schedule_policy	text
shift_match_mode	text
snapshot_schema	integer
published_at ?	timestamptz

DB-037
opportunity_skills

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK revision_id	DB-036
FK skill_id	DB-013
required	boolean
minimum_level ?	text

DB-038
opportunity_shifts

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK revision_id	DB-036
weekday	integer
start_minute	integer
end_minute	integer
timezone	text

DB-039
opportunity_hr_assignments

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK opportunity_id	DB-035
FK membership_id	DB-023
FK assigned_by_id	DB-001
accepted_at ?	timestamptz
revoked_at ?	timestamptz

DB-040
saved_opportunities

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK candidate_id	DB-012
FK opportunity_id	DB-035

APP | Ariza, suhbat va ish

Barcha maydonlar. FK yonidagi DB indeksi parent.idni bildiradi. Tashqi domainlar umumiy SVG/Mermaidda bog‘langan.

PK = primary key; ? = NULL; FK child -> parent. Parentda 0..N child, non-null FKda child uchun aynan1 parent.

DB-041
applications

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK candidate_id	DB-012
FK opportunity_id	DB-035
status	text
FK current_attempt_id ?	DB-042
version	integer
occupied_at ?	timestamptz
released_at ?	timestamptz

DB-042
application_attempts

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK application_id	DB-041
attempt_no	integer
FK revision_id	DB-036
FK enrollment_id ?	DB-015
profile_snapshot	jsonb
schedule_result	text
matching_version	text
proposed_slots ?	jsonb
submitted_at	timestamptz
idempotency_key	text

DB-043
application_events

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK application_id	DB-041
FK attempt_id ?	DB-042
FK actor_id	DB-001
from_status ?	text
to_status	text
reason ?	text
occurred_at	timestamptz

DB-044
interviews

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK application_id	DB-041
FK organizer_membership_id	DB-023
starts_at	timestamptz
ends_at	timestamptz
timezone	text
mode	text
location_or_link ?	text
status	text
version	integer

DB-045
interview_responses

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK interview_id	DB-044
FK user_id	DB-001
response	text
responded_at ?	timestamptz

DB-046
offers

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK application_id	DB-041
FK attempt_id	DB-042
FK revision_id	DB-036
FK issuer_membership_id	DB-023
terms_snapshot	jsonb
status	text
expires_at	timestamptz
accepted_at ?	timestamptz
idempotency_key	text

DB-047
job_engagements

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK application_id	DB-041
FK offer_id	DB-046
status	text
start_date	date
end_date ?	date
terms_snapshot	jsonb
confirmed_started_at ?	timestamptz

DB-048
engagement_confirmations

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK engagement_id	DB-047
FK user_id	DB-001
side	text
action	text
response	text
occurred_at	timestamptz

INT | Amaliyot va reja

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PK = primary key; ? = NULL; FK child -> parent. Parentda 0..N child, non-null FKda child uchun aynan1 parent.

DB-049
placements

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK application_id	DB-041
FK offer_id	DB-046
FK candidate_id	DB-012
FK organization_id	DB-021
FK enrollment_id ?	DB-015
academic_track	boolean
status	text
start_date	date
end_date	date
FK current_terms_id ?	DB-050
version	integer

DB-050
placement_terms

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK placement_id	DB-049
version	integer
schema_version	integer
terms_snapshot	jsonb
daily_limit_minutes	integer
weekly_limit_minutes	integer
required_minutes	integer
candidate_accepted_at ?	timestamptz
host_accepted_at ?	timestamptz

DB-051
mentor_assignments

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK placement_id	DB-049
FK membership_id	DB-023
FK assigned_by_id	DB-001
assigned_at	timestamptz
accepted_at ?	timestamptz
revoked_at ?	timestamptz

DB-052
academic_assignments

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK placement_id	DB-049
FK teacher_membership_id	DB-030
FK supervision_id	DB-034
FK assigned_by_id	DB-001
assigned_at	timestamptz
accepted_at ?	timestamptz
revoked_at ?	timestamptz

DB-053
plan_tasks

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK terms_id	DB-050
code	text
title	text
expected_result	text
required	boolean
target_minutes ?	integer

DB-054
placement_decisions

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK placement_id	DB-049
FK actor_id	DB-001
side	text
decision	text
FK terms_id	DB-050
reason ?	text
decided_at	timestamptz

LOG | Kundalik va dalillar

Barcha maydonlar. FK yonidagi DB indeksi parent.idni bildiradi. Tashqi domainlar umumiy SVG/Mermaidda bog‘langan.

PK = primary key; ? = NULL; FK child -> parent. Parentda 0..N child, non-null FKda child uchun aynan1 parent.

DB-055
daily_logs

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK placement_id	DB-049
local_date	date
FK current_revision_id ?	DB-056
version	integer

DB-056
log_revisions

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK daily_log_id	DB-055
revision	integer
FK terms_id	DB-050
FK author_id	DB-001
day_type	text
summary ?	text
total_minutes	integer
status	text
submitted_at ?	timestamptz
idempotency_key	text

DB-057
log_task_entries

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK revision_id	DB-056
FK plan_task_id ?	DB-053
result_text	text
minutes	integer
start_minute ?	integer
end_minute ?	integer

DB-058
log_evidence

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK entry_id	DB-057
FK file_id	DB-018
visibility	text

DB-059
workplace_log_reviews

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK revision_id	DB-056
FK mentor_assignment_id	DB-051
decision	text
comment ?	text
decided_at	timestamptz

DB-060
academic_log_reviews

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK revision_id	DB-056
FK academic_assignment_id	DB-052
decision	text
comment ?	text
decided_at	timestamptz

DB-061
log_amendment_requests

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK daily_log_id	DB-055
FK requester_id	DB-001
reason	text
status	text
FK reviewer_id ?	DB-001
resolved_at ?	timestamptz

RAT | Baholash va apellyatsiya

Barcha maydonlar. FK yonidagi DB indeksi parent.idni bildiradi. Tashqi domainlar umumiy SVG/Mermaidda bog‘langan.

PK = primary key; ? = NULL; FK child -> parent. Parentda 0..N child, non-null FKda child uchun aynan1 parent.

DB-062
rubric_versions

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
family	text
version	integer
status	text
scale_min	integer
scale_max	integer
published_at ?	timestamptz

DB-063
rubric_criteria

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK rubric_id	DB-062
code	text
title	text
weight	integer
required	boolean

DB-064
rating_submissions

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK placement_id ?	DB-049
FK engagement_id ?	DB-047
FK rubric_id	DB-062
FK rater_id	DB-001
FK target_user_id ?	DB-001
FK target_org_id ?	DB-021
version	integer
status	text
comment ?	text
submitted_at ?	timestamptz
publish_after ?	timestamptz

DB-065
rating_revisions

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK submission_id	DB-064
version	integer
comment ?	text
submitted_at	timestamptz
rubric_snapshot	jsonb

DB-066
rating_scores

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK rating_revision_id	DB-065
FK criterion_id	DB-063
score	integer
FK evidence_revision_id ?	DB-056

DB-067
rating_appeals

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK rating_revision_id	DB-065
FK appellant_id	DB-001
reason	text
status	text
FK reviewer_id ?	DB-001
decision ?	text
resolved_at ?	timestamptz

DB-068
rating_aggregates

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK target_user_id ?	DB-001
FK target_org_id ?	DB-021
FK rubric_id	DB-062
sample_size	integer
score ?	numeric(18,2)
window_start	date
window_end	date
computed_at	timestamptz

BIL | Tarif, hisob va to‘lov

Barcha maydonlar. FK yonidagi DB indeksi parent.idni bildiradi. Tashqi domainlar umumiy SVG/Mermaidda bog‘langan.

PK = primary key; ? = NULL; FK child -> parent. Parentda 0..N child, non-null FKda child uchun aynan1 parent.

DB-069
products

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
code	text
name	text
service_kind	text
active	boolean

DB-070
price_versions

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK product_id	DB-069
version	integer
amount	numeric(18,2)
currency	text
duration_days	integer
tax_included	boolean
valid_from	timestamptz
valid_to ?	timestamptz

DB-071
orders

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK organization_id	DB-021
FK purchaser_id	DB-001
FK price_id	DB-070
FK opportunity_id ?	DB-035
amount	numeric(18,2)
currency	text
price_snapshot	jsonb
status	text
idempotency_key	text

DB-072
invoices

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK order_id	DB-071
number	text
amount	numeric(18,2)
currency	text
fiscal_status	text
fiscal_reference ?	text
issued_at	timestamptz
paid_at ?	timestamptz

DB-073
payment_attempts

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK order_id	DB-071
provider	text
provider_transaction_id ?	text
amount	numeric(18,2)
currency	text
status	text
requested_at	timestamptz
confirmed_at ?	timestamptz

DB-074
payment_events

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK attempt_id	DB-073
provider	text
event_key	text
payload_digest	text
normalized_payload	jsonb
processed_at ?	timestamptz
received_at	timestamptz

DB-075
service_entitlements

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK organization_id	DB-021
FK order_id ?	DB-071
FK opportunity_id ?	DB-035
service_kind	text
status	text
starts_at ?	timestamptz
ends_at ?	timestamptz
quota ?	integer
FK granted_by_id ?	DB-001
grant_reason ?	text

DB-076
refunds

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK payment_attempt_id	DB-073
FK requester_id	DB-001
FK approver_id ?	DB-001
amount	numeric(18,2)
currency	text
reason	text
status	text
provider_refund_id ?	text
resolved_at ?	timestamptz

DB-077
billing_ledger

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK organization_id	DB-021
FK order_id ?	DB-071
FK payment_attempt_id ?	DB-073
FK refund_id ?	DB-076
entry_kind	text
amount	numeric(18,2)
currency	text
occurred_at	timestamptz
dedupe_key	text

OPS | Moderatsiya, yordam va maxfiylik

Barcha maydonlar. FK yonidagi DB indeksi parent.idni bildiradi. Tashqi domainlar umumiy SVG/Mermaidda bog‘langan.

PK = primary key; ? = NULL; FK child -> parent. Parentda 0..N child, non-null FKda child uchun aynan1 parent.

DB-078
moderation_cases

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK reporter_id	DB-001
FK opportunity_id ?	DB-035
FK organization_id ?	DB-021
FK rating_id ?	DB-064
FK file_id ?	DB-018
category	text
status	text
severity	text
FK assigned_to_id ?	DB-001
due_at	timestamptz
decision ?	text
decided_at ?	timestamptz

DB-079
moderation_actions

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK case_id	DB-078
FK actor_id	DB-001
action	text
reason	text
expires_at ?	timestamptz
FK appealed_action_id ?	DB-079
occurred_at	timestamptz

DB-080
support_tickets

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK requester_id	DB-001
category	text
priority	text
status	text
FK assignee_id ?	DB-001
subject	text
first_response_due_at	timestamptz
resolved_at ?	timestamptz

DB-081
support_messages

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK ticket_id	DB-080
FK author_id	DB-001
body	text
FK file_id ?	DB-018
internal	boolean

DB-082
privacy_requests

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK user_id	DB-001
kind	text
status	text
verified_at ?	timestamptz
due_at	timestamptz
FK result_file_id ?	DB-018
resolved_at ?	timestamptz
reason ?	text

DB-083
retention_policies

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
data_class	text
policy_version	integer
active_days ?	integer
backup_expiry_days	integer
legal_basis	text
status	text

DB-084
legal_holds

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK user_id ?	DB-001
FK organization_id ?	DB-021
FK case_id ?	DB-078
reason	text
FK authorized_by_id	DB-001
expires_at	timestamptz
released_at ?	timestamptz

SYS | Xabar, audit va boshqaruv

Barcha maydonlar. FK yonidagi DB indeksi parent.idni bildiradi. Tashqi domainlar umumiy SVG/Mermaidda bog‘langan.

PK = primary key; ? = NULL; FK child -> parent. Parentda 0..N child, non-null FKda child uchun aynan1 parent.

DB-085
notification_preferences

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK user_id	DB-001
channel	text
purpose	text
enabled	boolean
FK consent_event_id ?	DB-011

DB-086
outbox_events

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
event_type	text
schema_version	integer
aggregate_key	text
payload	jsonb
dedupe_key	text
available_at	timestamptz
processed_at ?	timestamptz
attempts	integer

DB-087
notifications

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK user_id	DB-001
FK event_id	DB-086
template_code	text
FK opportunity_id ?	DB-035
FK application_id ?	DB-041
FK placement_id ?	DB-049
FK ticket_id ?	DB-080
read_at ?	timestamptz

DB-088
notification_deliveries

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK notification_id	DB-087
channel	text
provider	text
provider_message_id ?	text
status	text
attempt_no	integer
next_retry_at ?	timestamptz
delivered_at ?	timestamptz
error_code ?	text

DB-089
audit_events

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK actor_id ?	DB-001
FK session_id ?	DB-004
action	text
object_type	text
object_key	text
scope_snapshot	jsonb
before_digest ?	text
after_digest ?	text
reason ?	text
request_id	text
occurred_at	timestamptz

DB-090
idempotency_keys

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
FK user_id	DB-001
operation	text
key	text
request_digest	text
response_status ?	integer
response_ref ?	text
expires_at	timestamptz

DB-091
analytics_events

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
event_key	text
event_name	text
actor_pseudonym ?	text
cohort ?	text
properties	jsonb
occurred_at	timestamptz
schema_version	integer

DB-092
feature_flags

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
code	text
enabled	boolean
rule_schema	jsonb
FK changed_by_id	DB-001
reason	text

DB-093
incidents

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
severity	text
status	text
FK commander_id	DB-001
summary	text
started_at	timestamptz
resolved_at ?	timestamptz
postmortem_ref ?	text

DB-094
restore_runs

PK id	uuid
created_at	timestamptz
updated_at	timestamptz
started_at	timestamptz
finished_at ?	timestamptz
backup_ref	text
status	text
recovered_through ?	timestamptz
measured_rpo_minutes ?	integer
measured_rto_minutes ?	integer
FK checked_by_id	DB-001
evidence_ref ?	text

USER -> IMKONIYAT -> BAZA | 1/4

Faol membership + rol + scope + amal + shaxsiy accept. Quyida DB indekslari, lug‘atda barcha maydonlar.

GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN	**CAP-001 | PUBLIC_SEARCH** E’lon qidirish MVP	DB-035 opportunities, DB-036 opportunity_revisions, DB-037 opportunity_skills, DB-038 opportunity_shifts, DB-021 organizations, DB-013 skills
GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN	**CAP-002 | AUTH_SELF** Hisob va kirish MVP	DB-001 users, DB-002 user_contacts, DB-003 auth_challenges, DB-004 sessions, DB-005 mfa_credentials, DB-006 recovery_requests, DB-011 consent_events
CANDIDATE, STUDENT	**CAP-003 | PROFILE_SELF** Profil va CV MVP	DB-012 candidate_profiles, DB-014 candidate_skills, DB-013 skills, DB-018 files
CANDIDATE, STUDENT	**CAP-004 | EDUCATION_SELF** Ta’lim tarixi va tasdiq MVP	DB-015 enrollments, DB-028 institutions, DB-019 verification_cases, DB-020 verification_evidence, DB-018 files
CANDIDATE, STUDENT	**CAP-005 | AVAILABILITY_SELF** Bo‘sh vaqt va istisno MVP	DB-016 availability_rules, DB-017 availability_exceptions
CANDIDATE, STUDENT	**CAP-006 | SAVE_JOB** E’lon saqlash MVP	DB-040 saved_opportunities, DB-035 opportunities
CANDIDATE, STUDENT	**CAP-007 | APPLY** Ariza va qayta ariza MVP	DB-041 applications, DB-042 application_attempts, DB-043 application_events, DB-036 opportunity_revisions, DB-090 idempotency_keys
OWNER, HR, HOST_COORDINATOR	**CAP-008 | APPLICATION_HR** Nomzodni ko‘rish va saralash MVP	DB-039 opportunity_hr_assignments, DB-041 applications, DB-042 application_attempts, DB-043 application_events, DB-018 files
CANDIDATE, STUDENT, OWNER, HR	**CAP-009 | INTERVIEW** Suhbat belgilash/javob MVP	DB-044 interviews, DB-045 interview_responses, DB-041 applications
OWNER, HR, HOST_COORDINATOR	**CAP-010 | OFFER_ISSUE** Taklif yaratish/bekor qilish MVP	DB-046 offers, DB-041 applications, DB-042 application_attempts

USER -> IMKONIYAT -> BAZA | 2/4

Faol membership + rol + scope + amal + shaxsiy accept. Quyida DB indekslari, lug‘atda barcha maydonlar.

CANDIDATE, STUDENT	**CAP-011 | OFFER_ACCEPT** Taklif qabul qilish MVP	DB-046 offers, DB-041 applications, DB-035 opportunities, DB-047 job_engagements, DB-049 placements, DB-090 idempotency_keys
CANDIDATE, STUDENT, OWNER, HR	**CAP-012 | JOB_CONFIRM** Ish boshlandi/yakun/nizo MVP	DB-047 job_engagements, DB-048 engagement_confirmations, DB-041 applications
OWNER, HOST_COORDINATOR	**CAP-013 | ORG_VERIFY** Tashkilot va birinchi vakil MVP	DB-021 organizations, DB-022 organization_capabilities, DB-019 verification_cases, DB-020 verification_evidence, DB-027 org_ownership_events
OWNER	**CAP-014 | STAFF_MANAGE** Xodim taklif va rol bekori MVP	DB-023 org_memberships, DB-024 org_role_grants, DB-025 org_invitations, DB-026 org_invitation_roles, DB-027 org_ownership_events
GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN	**CAP-015 | INVITE_ACCEPT** Xodim taklifini qabul MVP	DB-025 org_invitations, DB-026 org_invitation_roles, DB-023 org_memberships, DB-033 institution_invitations, DB-032 institution_appointments, DB-030 institution_memberships
OWNER, HR, HOST_COORDINATOR	**CAP-016 | JOB_PUBLISH** E’lon yaratish/nashr/tahrir MVP	DB-035 opportunities, DB-036 opportunity_revisions, DB-037 opportunity_skills, DB-038 opportunity_shifts, DB-039 opportunity_hr_assignments, DB-075 service_entitlements
OWNER, HOST_COORDINATOR	**CAP-017 | HOST_PLAN** Amaliyot reja va mentor MVP	DB-049 placements, DB-050 placement_terms, DB-053 plan_tasks, DB-051 mentor_assignments, DB-054 placement_decisions
CANDIDATE, STUDENT, MENTOR, TEACHER	**CAP-018 | PLACEMENT_ACCEPT** Amaliyot shartlari va assignment qabuli MVP	DB-049 placements, DB-050 placement_terms, DB-051 mentor_assignments, DB-052 academic_assignments, DB-054 placement_decisions
COORDINATOR	**CAP-019 | INSTITUTION_VERIFY** Rasmiy muassasa va tayinlov MVP	DB-028 institutions, DB-029 institution_agreements, DB-032 institution_appointments, DB-033 institution_invitations, DB-019 verification_cases
COORDINATOR	**CAP-020 | SUPERVISION_ASSIGN** Talaba va o‘qituvchi biriktirish MVP	DB-034 student_supervisions, DB-015 enrollments, DB-030 institution_memberships, DB-031 institution_role_grants, DB-052 academic_assignments

USER -> IMKONIYAT -> BAZA | 3/4

Faol membership + rol + scope + amal + shaxsiy accept. Quyida DB indekslari, lug‘atda barcha maydonlar.

COORDINATOR, TEACHER	**CAP-021 | ACADEMIC_VIEW** Akademik jarayonni kuzatish MVP	DB-034 student_supervisions, DB-052 academic_assignments, DB-049 placements, DB-055 daily_logs, DB-056 log_revisions
CANDIDATE, STUDENT	**CAP-022 | LOG_WRITE** Amaliyot kundaligi va dalil MVP	DB-055 daily_logs, DB-056 log_revisions, DB-057 log_task_entries, DB-058 log_evidence, DB-018 files, DB-061 log_amendment_requests
MENTOR	**CAP-023 | LOG_MENTOR** Ish joyi kundaligini tasdiqlash MVP	DB-059 workplace_log_reviews, DB-051 mentor_assignments, DB-055 daily_logs, DB-056 log_revisions, DB-057 log_task_entries, DB-058 log_evidence
TEACHER	**CAP-024 | LOG_ACADEMIC** Akademik fikr va yakun MVP	DB-060 academic_log_reviews, DB-052 academic_assignments, DB-056 log_revisions, DB-054 placement_decisions
CANDIDATE, STUDENT, HOST_COORDINATOR, MENTOR, TEACHER, COORDINATOR	**CAP-025 | PLACEMENT_CLOSE** Amaliyot yakuni/chiqish MVP	DB-054 placement_decisions, DB-049 placements, DB-050 placement_terms, DB-053 plan_tasks, DB-055 daily_logs
CANDIDATE, STUDENT, MENTOR, TEACHER, COORDINATOR	**CAP-026 | REPORT_EXPORT** Amaliyot hisobotini olish MVP	DB-049 placements, DB-055 daily_logs, DB-056 log_revisions, DB-059 workplace_log_reviews, DB-060 academic_log_reviews, DB-018 files, DB-089 audit_events
CANDIDATE, STUDENT, HR, HOST_COORDINATOR, MENTOR, TEACHER	**CAP-027 | RATING_SUBMIT** Munosabatdan keyin baho PILOT_FLAG	DB-062 rubric_versions, DB-063 rubric_criteria, DB-064 rating_submissions, DB-065 rating_revisions, DB-066 rating_scores
CANDIDATE, STUDENT, OWNER, HR, MENTOR, TEACHER, MODERATOR	**CAP-028 | RATING_APPEAL** Bahoga e’tiroz va agregat PILOT_FLAG	DB-067 rating_appeals, DB-064 rating_submissions, DB-065 rating_revisions, DB-068 rating_aggregates, DB-078 moderation_cases
OWNER, BILLING	**CAP-029 | BILLING** Tarif, order va hisob BILLING_GATE	DB-069 products, DB-070 price_versions, DB-071 orders, DB-072 invoices, DB-073 payment_attempts, DB-075 service_entitlements, DB-077 billing_ledger
OWNER, BILLING, FINANCE	**CAP-030 | REFUND** Pul qaytarish va reconciliation BILLING_GATE	DB-076 refunds, DB-073 payment_attempts, DB-074 payment_events, DB-077 billing_ledger, DB-071 orders

USER -> IMKONIYAT -> BAZA | 4/4

Faol membership + rol + scope + amal + shaxsiy accept. Quyida DB indekslari, lug‘atda barcha maydonlar.

CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN	**CAP-031 | NOTIFICATION** Xabar va kanal sozlash MVP	DB-085 notification_preferences, DB-087 notifications, DB-088 notification_deliveries, DB-086 outbox_events, DB-011 consent_events
CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, TEACHER, COORDINATOR	**CAP-032 | REPORT_ABUSE** Shikoyat va apellyatsiya MVP	DB-078 moderation_cases, DB-079 moderation_actions, DB-018 files
MODERATOR	**CAP-033 | MODERATE** Tekshirish/moderatsiya MVP	DB-019 verification_cases, DB-020 verification_evidence, DB-078 moderation_cases, DB-079 moderation_actions, DB-021 organizations, DB-035 opportunities
CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN	**CAP-034 | SUPPORT** Yordam murojaati MVP	DB-080 support_tickets, DB-081 support_messages, DB-018 files
CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN	**CAP-035 | PRIVACY** Rozilik, eksport, tuzatish/o‘chirish MVP	DB-082 privacy_requests, DB-083 retention_policies, DB-084 legal_holds, DB-011 consent_events, DB-018 files
ADMIN	**CAP-036 | ADMIN_AUDIT** Audit va vaqtli admin kirish MVP	DB-089 audit_events, DB-010 platform_role_grants, DB-009 role_permissions, DB-008 role_catalog, DB-007 permission_catalog, DB-004 sessions, DB-093 incidents
ADMIN	**CAP-037 | OPERATIONS** Flag, incident, backup va o‘lchov MVP	DB-092 feature_flags, DB-093 incidents, DB-094 restore_runs, DB-091 analytics_events, DB-086 outbox_events
OWNER, BILLING, FINANCE	**CAP-038 | FINANCE_REPORT** Moliyaviy hisobot BILLING_GATE	DB-077 billing_ledger, DB-071 orders, DB-072 invoices, DB-075 service_entitlements
GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN	**CAP-039 | PUBLIC_INFO** Ochiq xizmat va siyosat ma’lumoti MVP	DB-069 products, DB-070 price_versions, DB-011 consent_events, DB-092 feature_flags
GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR	**CAP-040 | RATING_PUBLIC** Korxonaning ochiq reytingi PILOT_FLAG	DB-068 rating_aggregates, DB-062 rubric_versions, DB-021 organizations, DB-092 feature_flags

To‘liq ma’lumotlar bazasi lug‘ati

Bu mantiqiy/texnik model. **?** NULL mumkinligini anglatadi. Har FK parent.idga; parentda 0..N child bo‘lishi mumkin, childda nullable bo‘lmasa aynan1 parent. Birga UNIQUE bo‘lgan FK jufti1:1 yoki shu munosabatning bitta nusxasini cheklaydi. Jadvallar ko‘pligi modulli monolitni mikroservisga aylantirmaydi.

IAM - Hisob, kirish va ruxsatlar

DB-001 users

Bitta shaxsning asosiy hisobi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
display_name	text / NO	-
status	text / NO	-
password_hash	text / YES	-
locale	text / NO	-
timezone	text / NO	-
adult_declared_at	timestamptz / YES	-
last_login_at	timestamptz / YES	-

CHECK: status IN ('ACTIVE','LIMITED','SUSPENDED','DELETION_PENDING','ANONYMIZED')

INDEX: status,created_at

Parol xeshi framework orqali; ochiq parol saqlanmaydi. Tahrir updated_at server vaqti bilan.

Imkoniyatlar: CAP-002.

DB-002 user_contacts

Tasdiqlangan yoki kutilayotgan login kontaktlari.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
user_id	uuid / NO	DB-001 users.id
channel	text / NO	-

Maydon	Type / NULL	PK yoki FK
normalized_value	text / NO	-
verified_at	timestamptz / YES	-
revoked_at	timestamptz / YES	-
is_primary	boolean / NO	-

UNIQUE: channel,normalized_value

CHECK: channel IN ('EMAIL','PHONE')

INDEX: user_id,channel

WhatsApp va SMS bir PHONE kontaktining yetkazish kanali. Revoked kontakt qayta berilishi faqat tekshirilgan recovery bilan; avtomatik account merge yo‘q.

Imkoniyatlar: CAP-002.

DB-003 auth_challenges

Maqsadga bog‘langan bir martalik kod.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
user_id	uuid / YES	DB-001 users.id
contact_id	uuid / YES	DB-002 user_contacts.id
destination_digest	text / NO	-
channel	text / NO	-
purpose	text / NO	-
code_hmac	text / NO	-
expires_at	timestamptz / NO	-
attempts	integer / NO	-
consumed_at	timestamptz / YES	-
superseded_at	timestamptz / YES	-

CHECK: attempts >= 0 AND attempts <= 5

INDEX: destination_digest,purpose,created_at; expires_at

Kontakt hali mavjud bo‘lmasa digest ishlaydi; raw manzil faqat qisqa muddatli shifrlangan delivery payload. Har consume atomar va bir martalik.

Imkoniyatlar: CAP-002.

DB-004 sessions

Qurilma va sessiyani bekor qilish.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
user_id	uuid / NO	DB-001 users.id
token_digest	text / NO	-
device_key_digest	text / YES	-
assurance	text / NO	-
expires_at	timestamptz / NO	-
revoked_at	timestamptz / YES	-
last_seen_at	timestamptz / NO	-

UNIQUE: token_digest

INDEX: user_id,revoked_at; expires_at

Cookie HttpOnly/Secure; yangi qurilmada PHONE-only eski ma’lumotni ochmaydi. Revoke barcha sezgir so‘rovlarda tekshiriladi.

Imkoniyatlar: CAP-002, CAP-036.

DB-005 mfa_credentials

TOTP/passkey va tiklash omili.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
user_id	uuid / NO	DB-001 users.id
kind	text / NO	-
credential_key	text / NO	-
secret_encrypted	text / YES	-
public_key	text / YES	-
sign_count	integer / NO	-
verified_at	timestamptz / YES	-

Maydon	Type / NULL	PK yoki FK
revoked_at	timestamptz / YES	-

UNIQUE: credential_key

INDEX: user_id,revoked_at

TOTP siri kalit boshqaruvi bilan shifrlanadi. Passkey private key serverda bo‘lmaydi; recovery kodlar alohida xeshlangan qiymatlar.

Imkoniyatlar: CAP-002.

DB-006 recovery_requests

Kontakt, dublikat va oxirgi egani tiklash.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
user_id	uuid / YES	DB-001 users.id
kind	text / NO	-
status	text / NO	-
requester_contact_digest	text / NO	-
reviewer_id	uuid / YES	DB-001 users.id
evidence_file_id	uuid / YES	DB-018 files.id
resolved_at	timestamptz / YES	-

INDEX: status,created_at

Reviewer requester bo‘la olmaydi. Account merge FKlarni auditlangan migratsiya bilan ko‘chiradi; uchinchi tomon huquqlarini qo‘shib yubormaydi.

Imkoniyatlar: CAP-002.

DB-007 permission_catalog

Atomar imkoniyat kodi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
code	text / NO	-

Maydon	Type / NULL	PK yoki FK
description	text / NO	-

UNIQUE: code

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Imkoniyatlar: CAP-036.

DB-008 role_catalog

Rol va uning ruxsat doirasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
code	text / NO	-
scope_kind	text / NO	-
description	text / NO	-

UNIQUE: code

CHECK: scope_kind IN ('SELF','ORG','INSTITUTION','PLATFORM')

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Imkoniyatlar: CAP-036.

DB-009 role_permissions

Rol-imkoniyat bog‘lanishi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
role_id	uuid / NO	DB-008 role_catalog.id
permission_id	uuid / NO	DB-007 permission_catalog.id

UNIQUE: role_id,permission_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Imkoniyatlar: CAP-036.

DB-010 platform_role_grants

Operatorning vaqtli vakolati.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
user_id	uuid / NO	DB-001 users.id
role_id	uuid / NO	DB-008 role_catalog.id
granted_by_id	uuid / NO	DB-001 users.id
expires_at	timestamptz / NO	-
revoked_at	timestamptz / YES	-

INDEX: user_id,expires_at

Faqat PLATFORM role; grantor o‘ziga grant bera olmaydi. Sezgir grant ikki mustaqil tasdiq va audit bilan.

Imkoniyatlar: CAP-036.

DB-011 consent_events

Rozilik va huquqiy asos tarixi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
user_id	uuid / NO	DB-001 users.id
purpose	text / NO	-
policy_version	text / NO	-
action	text / NO	-
legal_basis	text / NO	-
occurred_at	timestamptz / NO	-

INDEX: user_id,purpose,occurred_at

Append-only. Marketing opt-in alohida; xizmat uchun zarur ishlovga majburiy marketing ulanmaydi.

Imkoniyatlar: CAP-002, CAP-031, CAP-035, CAP-039.

CAN - Nomzod, malaka va vaqt

DB-012 candidate_profiles

Talaba bo‘lmagan va talaba nomzodning umumiy profili.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
user_id	uuid / NO	DB-001 users.id
headline	text / NO	-
region	text / NO	-
bio	text / YES	-
experience_months	integer / NO	-
visibility	text / NO	-
cv_file_id	uuid / YES	DB-018 files.id

UNIQUE: user_id

CHECK: experience_months >= 0

INDEX: region

Talabalik bu jadvalning majburiy ustuni emas. CV faqat nomzod va arizasi bo‘yicha ruxsatli HRga; global ochiq eksport yo‘q.

Imkoniyatlar: CAP-003.

DB-013 skills

UZ/RU qidiruv uchun malaka katalogi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
code	text / NO	-
name_uz	text / NO	-
name_ru	text / NO	-
aliases	jsonb / NO	-
active	boolean / NO	-

UNIQUE: code

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Imkoniyatlar: CAP-001, CAP-003.

DB-014 candidate_skills

Malaka va tasdiq manbasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
candidate_id	uuid / NO	DB-012 candidate_profiles.id
skill_id	uuid / NO	DB-013 skills.id
level	text / NO	-
verification_case_id	uuid / YES	DB-019 verification_cases.id

UNIQUE: candidate_id,skill_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Self-declared va tasdiqlangan malaka farqlanadi; case APPROVED bo‘lmasa verified ko‘rinmaydi.

Imkoniyatlar: CAP-003.

DB-015 enrollments

Muassasaga oid ta’lim tarixi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
candidate_id	uuid / NO	DB-012 candidate_profiles.id
institution_id	uuid / NO	DB-028 institutions.id
program	text / NO	-
student_ref_digest	text / YES	-
start_date	date / NO	-
end_date	date / YES	-
status	text / NO	-
verification_case_id	uuid / YES	DB-019 verification_cases.id

CHECK: end_date IS NULL OR end_date >= start_date

INDEX: candidate_id,status; institution_id,status

Tarix saqlanadi. Bir nechta ta’lim bo‘lishi mumkin; biriktirish aniq enrollmentga. Tasdiq muddati bitganda umumiy ish huquqi qoladi.

Imkoniyatlar: CAP-004, CAP-020.

DB-016 availability_rules

Haftalik mavjudlik.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
candidate_id	uuid / NO	DB-012 candidate_profiles.id
weekday	integer / NO	-
start_minute	integer / NO	-
end_minute	integer / NO	-
valid_from	date / NO	-
valid_to	date / YES	-
timezone	text / NO	-

CHECK: weekday BETWEEN 1 AND 7; start_minute >= 0 AND end_minute <= 1440 AND start_minute < end_minute; valid_to IS NULL OR valid_to >= valid_from

INDEX: candidate_id,weekday,valid_from

Oraliqlar [start,end). Kesishgan mavjudlik union qilinadi, daqiqalar ikki marta sanalmaydi.

Imkoniyatlar: CAP-005.

DB-017 availability_exceptions

Sana bo‘yicha dars/istisno.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
candidate_id	uuid / NO	DB-012 candidate_profiles.id
local_date	date / NO	-
kind	text / NO	-

Maydon	Type / NULL	PK yoki FK
start_minute	integer / NO	-
end_minute	integer / NO	-

CHECK: kind IN ('AVAILABLE','UNAVAILABLE'); start_minute >= 0 AND end_minute <= 1440 AND start_minute < end_minute

INDEX: candidate_id,local_date

Imkoniyatlar: CAP-005.

DB-018 files

Xususiy, skanerdan o‘tadigan fayl.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
owner_user_id	uuid / NO	DB-001 users.id
storage_key	text / NO	-
filename	text / NO	-
mime_type	text / NO	-
byte_size	integer / NO	-
sha256	text / NO	-
scan_status	text / NO	-
retention_class	text / NO	-
deleted_at	timestamptz / YES	-

UNIQUE: storage_key

CHECK: byte_size > 0

INDEX: owner_user_id,created_at; scan_status,created_at

Baza fayl tanasini saqlamaydi. Karantin, MIME/magic tekshiruvi, hajm limiti, qisqa signed URL va yuklash huquqi shart. Maxfiylikni key nomi bilan hal qilib bo‘lmaydi.

Imkoniyatlar: CAP-003, CAP-004, CAP-008, CAP-022, CAP-026, CAP-032, CAP-034, CAP-035.

DB-019 verification_cases

Shaxs, tashkilot yoki ta’limni tekshirish navbati.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
requester_id	uuid / NO	DB-001 users.id
candidate_id	uuid / YES	DB-012 candidate_profiles.id
organization_id	uuid / YES	DB-021 organizations.id
institution_id	uuid / YES	DB-028 institutions.id
enrollment_id	uuid / YES	DB-015 enrollments.id
appointment_id	uuid / YES	DB-032 institution_appointments.id
kind	text / NO	-
status	text / NO	-
current_revision	integer / NO	-
assigned_reviewer_id	uuid / YES	DB-001 users.id
due_at	timestamptz / NO	-
decision_reason	text / YES	-
decided_at	timestamptz / YES	-

CHECK: num_nonnulls(candidate_id,organization_id,institution_id,enrollment_id,appointment_id) = 1; current_revision > 0

INDEX: status,due_at; assigned_reviewer_id,status

Har tur uchun haqiqiy FK, target_type/target_id jufti yo‘q. Tekshiruvchi o‘z murojaatini tasdiqlamaydi.

Imkoniyatlar: CAP-004, CAP-013, CAP-019, CAP-033.

DB-020 verification_evidence

Tekshiruvga topshirilgan revision dalili.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
case_id	uuid / NO	DB-019 verification_cases.id

Maydon	Type / NULL	PK yoki FK
revision	integer / NO	-
file_id	uuid / YES	DB-018 files.id
note	text / YES	-

CHECK: revision > 0; file_id IS NOT NULL OR note IS NOT NULL

INDEX: case_id,revision

Imkoniyatlar: CAP-004, CAP-013, CAP-033.

ORG - Tashkilot va xodimlar

DB-021 organizations

STIR bo‘yicha yagona ish beruvchi/host.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
tax_id	text / NO	-
legal_name	text / NO	-
public_name	text / NO	-
region	text / NO	-
description	text / YES	-
website	text / YES	-
status	text / NO	-

UNIQUE: tax_id

INDEX: status,region

Mavjud STIR yangi tashkilot yaratmaydi. Birinchi vakil mustaqil tekshiruvdan so‘ng; public ma’lumot va maxfiy dalil ajratiladi.

Imkoniyatlar: CAP-001, CAP-013, CAP-033, CAP-040.

DB-022 organization_capabilities

Ish e’loni va host huquqi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK

Maydon	Type / NULL	PK yoki FK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
organization_id	uuid / NO	DB-021 organizations.id
capability	text / NO	-
status	text / NO	-
approved_case_id	uuid / YES	DB-019 verification_cases.id

UNIQUE: organization_id,capability

CHECK: capability IN ('JOB_EMPLOYER','INTERNSHIP_HOST')

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Tasdiqlangan capability ham faol a’zo rolini almashtirmaydi.

Imkoniyatlar: CAP-013.

DB-023 org_memberships

Xodimning tashkilotga mansubligi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
organization_id	uuid / NO	DB-021 organizations.id
user_id	uuid / NO	DB-001 users.id
status	text / NO	-
accepted_at	timestamptz / YES	-
revoked_at	timestamptz / YES	-

UNIQUE: organization_id,user_id; id,organization_id

INDEX: user_id,status

Ownerdan boshqa a’zolik pul va egalik huquqini avtomatik bermaydi. Oxirgi active owner tranzaksiyada himoyalanadi.

Imkoniyatlar: CAP-014, CAP-015.

DB-024 org_role_grants

Tashkilot ichidagi rol.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
membership_id	uuid / NO	DB-023 org_memberships.id
role_id	uuid / NO	DB-008 role_catalog.id
granted_by_id	uuid / NO	DB-001 users.id
revoked_at	timestamptz / YES	-

UNIQUE: membership_id,role_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Faqat ORG role. Ruxsat aktif membership, role va assignment kesishmasi; global union emas.

Imkoniyatlar: CAP-014.

DB-025 org_invitations

Aniq kontaktga vaqtli taklif.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
organization_id	uuid / NO	DB-021 organizations.id
inviter_membership_id	uuid / NO	DB-023 org_memberships.id
contact_digest	text / NO	-
token_digest	text / NO	-
expires_at	timestamptz / NO	-
accepted_by_id	uuid / YES	DB-001 users.id
accepted_at	timestamptz / YES	-
revoked_at	timestamptz / YES	-

UNIQUE: token_digest

INDEX: organization_id,expires_at

24 soat; inviter ayni orgga mansub; kontakt va qabul qiluvchi mos; repeat accept idempotent.

Imkoniyatlar: CAP-014, CAP-015.

DB-026 org_invitation_roles

Taklifdagi ochiq ko‘rsatilgan rol.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
invitation_id	uuid / NO	DB-025 org_invitations.id
role_id	uuid / NO	DB-008 role_catalog.id

UNIQUE: invitation_id,role_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Imkoniyatlar: CAP-014, CAP-015.

DB-027 org_ownership_events

Birinchi va keyingi egani tayinlash dalili.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
organization_id	uuid / NO	DB-021 organizations.id
new_membership_id	uuid / NO	DB-023 org_memberships.id
previous_membership_id	uuid / YES	DB-023 org_memberships.id
verification_case_id	uuid / NO	DB-019 verification_cases.id
approved_by_id	uuid / NO	DB-001 users.id
effective_at	timestamptz / NO	-

INDEX: organization_id,effective_at

Grant mustaqil reviewer bilan; oxirgi egani yo‘qotishdan oldin o‘rinbosar yoki recovery. Append-only.

Imkoniyatlar: CAP-013, CAP-014.

EDU - Muassasa va rasmiy biriktirish

DB-028 institutions

Hamkor o‘quv muassasasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
registry_code	text / NO	-
name	text / NO	-
region	text / NO	-
status	text / NO	-

UNIQUE: registry_code

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Imkoniyatlar: CAP-004, CAP-019.

DB-029 institution_agreements

Akademik xizmatning rasmiy chegarasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
institution_id	uuid / NO	DB-028 institutions.id
version	integer / NO	-
status	text / NO	-
valid_from	date / NO	-
valid_to	date / YES	-
purposes	jsonb / NO	-
signed_file_id	uuid / YES	DB-018 files.id

UNIQUE: institution_id,version

CHECK: version > 0; valid_to IS NULL OR valid_to >= valid_from

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Rasmiy baho, imzo, eksport va ishlov asosi kelishuvga bog‘liq; umumiy platforma PDFsi imzo deb taqdim etilmaydi.

Imkoniyatlar: CAP-019.

DB-030 institution_memberships

Rasmiy xodim mansubligi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
institution_id	uuid / NO	DB-028 institutions.id
user_id	uuid / NO	DB-001 users.id
status	text / NO	-
accepted_at	timestamptz / YES	-
revoked_at	timestamptz / YES	-

UNIQUE: institution_id,user_id; id,institution_id

INDEX: user_id,status

Imkoniyatlar: CAP-015, CAP-020.

DB-031 institution_role_grants

Koordinator/o‘qituvchi roli.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
membership_id	uuid / NO	DB-030 institution_memberships.id
role_id	uuid / NO	DB-008 role_catalog.id
appointment_id	uuid / NO	DB-032 institution_appointments.id
revoked_at	timestamptz / YES	-

UNIQUE: membership_id,role_id,appointment_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Faqat INSTITUTION role va ayni muassasaning faol rasmiy appointmenti bilan.

Imkoniyatlar: CAP-020.

DB-032 institution_appointments

Muassasa topshirig‘i va dalili.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
institution_id	uuid / NO	DB-028 institutions.id
user_id	uuid / NO	DB-001 users.id
document_file_id	uuid / NO	DB-018 files.id
status	text / NO	-
verification_case_id	uuid / YES	DB-019 verification_cases.id
valid_from	date / NO	-
valid_to	date / YES	-
accepted_at	timestamptz / YES	-

CHECK: valid_to IS NULL OR valid_to >= valid_from

INDEX: institution_id,status

Email domeni o‘zi tayinlov emas. Tayinlangan shaxs taklifni qabul qiladi, moderator o‘zini tayinlamaydi.

Imkoniyatlar: CAP-015, CAP-019.

DB-033 institution_invitations

Muassasa tayinlovi taklifi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
appointment_id	uuid / NO	DB-032 institution_appointments.id
contact_digest	text / NO	-
token_digest	text / NO	-
expires_at	timestamptz / NO	-
accepted_by_id	uuid / YES	DB-001 users.id
accepted_at	timestamptz / YES	-
revoked_at	timestamptz / YES	-

UNIQUE: token_digest

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Imkoniyatlar: CAP-015, CAP-019.

DB-034 student_supervisions

Koordinatorning aniq ta’lim yozuviga doirasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
enrollment_id	uuid / NO	DB-015 enrollments.id
coordinator_membership_id	uuid / NO	DB-030 institution_memberships.id
agreement_id	uuid / NO	DB-029 institution_agreements.id
purpose	text / NO	-
status	text / NO	-
legal_basis	text / NO	-
accepted_at	timestamptz / YES	-
revoked_at	timestamptz / YES	-

UNIQUE: enrollment_id,coordinator_membership_id,purpose

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Uch yozuvning institution_id qiymati teng. O‘quvchi boshqa muassasaga o‘tsa eski scope ko‘chmaydi; mas’ulga berilgan maqsad bilangina ko‘rinadi.

Imkoniyatlar: CAP-020, CAP-021.

JOB - E’lon va qidiruv

DB-035 opportunities

E’lon identifikatori va band o‘rin.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
organization_id	uuid / NO	DB-021 organizations.id

Maydon	Type / NULL	PK yoki FK
kind	text / NO	-
status	text / NO	-
current_revision_id	uuid / YES	DB-036 opportunity_revisions.id
capacity	integer / NO	-
occupied_count	integer / NO	-
published_at	timestamptz / YES	-
expires_at	timestamptz / YES	-
closed_at	timestamptz / YES	-

CHECK: kind IN ('PART_TIME','INTERNSHIP'); capacity > 0; occupied_count >= 0 AND occupied_count <= capacity

INDEX: status,published_at,id; organization_id,status

Counter va offer qabuli bir tranzaksiya/lockda. Active occupied applications bilan kunlik reconciliation; counter mustaqil tahrirlanmaydi.

Imkoniyatlar: CAP-001, CAP-006, CAP-011, CAP-016, CAP-033.

DB-036 opportunity_revisions

E’lon shartlarining o‘zgarmas nusxasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
opportunity_id	uuid / NO	DB-035 opportunities.id
revision	integer / NO	-
author_id	uuid / NO	DB-001 users.id
title	text / NO	-
description	text / NO	-
region	text / NO	-
work_mode	text / NO	-
eligibility	text / NO	-
program_required	text / YES	-
salary_min	numeric(18,2) / YES	-
salary_max	numeric(18,2) / YES	-

Maydon	Type / NULL	PK yoki FK
currency	text / NO	-
salary_period	text / YES	-
salary_basis	text / YES	-
is_paid	boolean / NO	-
start_date	date / NO	-
end_date	date / YES	-
schedule_policy	text / NO	-
shift_match_mode	text / NO	-
snapshot_schema	integer / NO	-
published_at	timestamptz / YES	-

UNIQUE: opportunity_id,revision; id,opportunity_id

CHECK: revision > 0; end_date IS NULL OR end_date >= start_date; salary_min IS NULL OR salary_min > 0; salary_max IS NULL OR salary_max >= salary_min

INDEX: opportunity_id,revision

PART_TIME va pullik INTERNSHIP uchun salary_min/max, UZS, davr/brutto-netto; INTERNSHIP end majburiy; unpaidda salary null. Bu cross-table qoidalar publish tranzaksiyasi va acceptance test bilan. Nashrdagi qator update qilinmaydi.

Imkoniyatlar: CAP-001, CAP-007, CAP-016.

DB-037 opportunity_skills

E’lon revisionining malaka talabi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
revision_id	uuid / NO	DB-036 opportunity_revisions.id
skill_id	uuid / NO	DB-013 skills.id
required	boolean / NO	-
minimum_level	text / YES	-

UNIQUE: revision_id,skill_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Imkoniyatlar: CAP-001, CAP-016.

DB-038 opportunity_shifts

E’lon revisioni smenalari.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
revision_id	uuid / NO	DB-036 opportunity_revisions.id
weekday	integer / NO	-
start_minute	integer / NO	-
end_minute	integer / NO	-
timezone	text / NO	-

CHECK: weekday BETWEEN 1 AND 7; start_minute >= 0 AND end_minute <= 1440 AND start_minute < end_minute

INDEX: revision_id,weekday

Tungi smena ikki sana/hafta kuniga bo‘linadi; barcha intervallar yarim ochiq.

Imkoniyatlar: CAP-001, CAP-016.

DB-039 opportunity_hr_assignments

HRning aynan e’longa doirasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
opportunity_id	uuid / NO	DB-035 opportunities.id
membership_id	uuid / NO	DB-023 org_memberships.id
assigned_by_id	uuid / NO	DB-001 users.id
accepted_at	timestamptz / YES	-
revoked_at	timestamptz / YES	-

UNIQUE: opportunity_id,membership_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

E’lon va membership tashkiloti teng, HR roli aktiv. Organization owner ham ko‘rish amalida shu maqsad va audit talabiga bo‘ysunadi.

Imkoniyatlar: CAP-008, CAP-016.

DB-040 saved_opportunities

Nomzod saqlagan e’lon.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
candidate_id	uuid / NO	DB-012 candidate_profiles.id
opportunity_id	uuid / NO	DB-035 opportunities.id

UNIQUE: candidate_id,opportunity_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Imkoniyatlar: CAP-006.

APP - Ariza, suhbat va ish

DB-041 applications

Bitta nomzod va bitta e’lon munosabati.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
candidate_id	uuid / NO	DB-012 candidate_profiles.id
opportunity_id	uuid / NO	DB-035 opportunities.id
status	text / NO	-
current_attempt_id	uuid / YES	DB-042 application_attempts.id
version	integer / NO	-
occupied_at	timestamptz / YES	-
released_at	timestamptz / YES	-

UNIQUE: candidate_id,opportunity_id; id,candidate_id,opportunity_id

CHECK: version > 0

INDEX: opportunity_id,status,created_at; candidate_id,created_at

Retry yangi application yaratmaydi. Version conditional update, transition va capacity bir tranzaksiyada.

Imkoniyatlar: CAP-007, CAP-008, CAP-009, CAP-010, CAP-011, CAP-012.

DB-042 application_attempts

Qayta ariza va o‘sha paytdagi shartlar.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
application_id	uuid / NO	DB-041 applications.id
attempt_no	integer / NO	-
revision_id	uuid / NO	DB-036 opportunity_revisions.id
enrollment_id	uuid / YES	DB-015 enrollments.id
profile_snapshot	jsonb / NO	-
schedule_result	text / NO	-
matching_version	text / NO	-
proposed_slots	jsonb / YES	-
submitted_at	timestamptz / NO	-
idempotency_key	text / NO	-

UNIQUE: application_id,attempt_no; idempotency_key

CHECK: attempt_no > 0

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Revision ayni e’longa, enrollment ayni candidatega tegishli. Snapshot schema-valid va minimallashtirilgan; keyinchalik CV o‘zgarishi tarixni buzmaydi.

Imkoniyatlar: CAP-007, CAP-008, CAP-010.

DB-043 application_events

Holat o‘zgarishi va qayta ochish.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
application_id	uuid / NO	DB-041 applications.id
attempt_id	uuid / YES	DB-042 application_attempts.id
actor_id	uuid / NO	DB-001 users.id

Maydon	Type / NULL	PK yoki FK
from_status	text / YES	-
to_status	text / NO	-
reason	text / YES	-
occurred_at	timestamptz / NO	-

INDEX: application_id,occurred_at

Append-only. REJECTED/EXPIREDdan qayta ochish HR ruxsati bilan; 24h va 3/30kun limit service tranzaksiyasida.

Imkoniyatlar: CAP-007, CAP-008.

DB-044 interviews

Suhbat va uning jadvali.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
application_id	uuid / NO	DB-041 applications.id
organizer_membership_id	uuid / NO	DB-023 org_memberships.id
starts_at	timestamptz / NO	-
ends_at	timestamptz / NO	-
timezone	text / NO	-
mode	text / NO	-
location_or_link	text / YES	-
status	text / NO	-
version	integer / NO	-

CHECK: ends_at > starts_at; version > 0

INDEX: application_id,starts_at

Tashkilot doirasi tekshiriladi; o‘zgarishlar audit/eventga. MVP platformadagi tasdiq, tashqi calendar sync emas.

Imkoniyatlar: CAP-009.

DB-045 interview_responses

Suhbatga tomonlarning javobi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
interview_id	uuid / NO	DB-044 interviews.id
user_id	uuid / NO	DB-001 users.id
response	text / NO	-
responded_at	timestamptz / YES	-

UNIQUE: interview_id,user_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Imkoniyatlar: CAP-009.

DB-046 offers

Muddati va shartlari aniq taklif.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
application_id	uuid / NO	DB-041 applications.id
attempt_id	uuid / NO	DB-042 application_attempts.id
revision_id	uuid / NO	DB-036 opportunity_revisions.id
issuer_membership_id	uuid / NO	DB-023 org_memberships.id
terms_snapshot	jsonb / NO	-
status	text / NO	-
expires_at	timestamptz / NO	-
accepted_at	timestamptz / YES	-
idempotency_key	text / NO	-

UNIQUE: idempotency_key

INDEX: application_id,status; status,expires_at

Bir applicationda bir ACTIVE offer uchun partial unique index. OFFERED joyni kafolatlamaydi; ACCEPT atomar capacity tekshiradi; expires_at serverda ham tekshiriladi.

Imkoniyatlar: CAP-010, CAP-011.

DB-047 job_engagements

Oddiy maoshli ishning boshlanish/yakun holati.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
application_id	uuid / NO	DB-041 applications.id
offer_id	uuid / NO	DB-046 offers.id
status	text / NO	-
start_date	date / NO	-
end_date	date / YES	-
terms_snapshot	jsonb / NO	-
confirmed_started_at	timestamptz / YES	-

UNIQUE: application_id; offer_id

CHECK: end_date IS NULL OR end_date >= start_date

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Faqat PART_TIME. Ikki tomon STARTni tasdiqlamaguncha haqiqiy ishga joylashish statistikasi emas; kundalik avtomatik ochilmaydi.

Imkoniyatlar: CAP-011, CAP-012.

DB-048 engagement_confirmations

Ish munosabatining ikki tomon tasdig‘i.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
engagement_id	uuid / NO	DB-047 job_engagements.id
user_id	uuid / NO	DB-001 users.id
side	text / NO	-

Maydon	Type / NULL	PK yoki FK
action	text / NO	-
response	text / NO	-
occurred_at	timestamptz / NO	-

UNIQUE: engagement_id,side,action

CHECK: side IN ('CANDIDATE','EMPLOYER')

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Nomzod o‘zi yoki berilgan vakolatli HR; side foydalanuvchi tanlagan erkin satrga ishonib berilmaydi.

Imkoniyatlar: CAP-012.

INT - Amaliyot va reja

DB-049 placements

Qabul qilingan amaliyot munosabati.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
application_id	uuid / NO	DB-041 applications.id
offer_id	uuid / NO	DB-046 offers.id
candidate_id	uuid / NO	DB-012 candidate_profiles.id
organization_id	uuid / NO	DB-021 organizations.id
enrollment_id	uuid / YES	DB-015 enrollments.id
academic_track	boolean / NO	-
status	text / NO	-
start_date	date / NO	-
end_date	date / NO	-
current_terms_id	uuid / YES	DB-050 placement_terms.id
version	integer / NO	-

UNIQUE: application_id; offer_id

CHECK: end_date >= start_date; NOT academic_track OR enrollment_id IS NOT NULL; version > 0

INDEX: candidate_id,status; organization_id,status

Faqat INTERNSHIP. Nomzod/application/offer/host bir-biriga mos. Academic ACTIVEda faol agreement, supervisor va mentor qabuli kerak.

Imkoniyatlar: CAP-011, CAP-017, CAP-018, CAP-021, CAP-025, CAP-026.

DB-050 placement_terms

Amaliyot rejasining immutable versiyasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
placement_id	uuid / NO	DB-049 placements.id
version	integer / NO	-
schema_version	integer / NO	-
terms_snapshot	jsonb / NO	-
daily_limit_minutes	integer / NO	-
weekly_limit_minutes	integer / NO	-
required_minutes	integer / NO	-
candidate_accepted_at	timestamptz / YES	-
host_accepted_at	timestamptz / YES	-

UNIQUE: placement_id,version; id,placement_id

CHECK: version > 0; daily_limit_minutes > 0 AND daily_limit_minutes <= 1440; weekly_limit_minutes > 0 AND weekly_limit_minutes <= 10080; required_minutes >= 0

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Imkoniyatlar: CAP-017, CAP-018, CAP-025.

DB-051 mentor_assignments

Korxona ustozi shaxsiy qabuli.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
placement_id	uuid / NO	DB-049 placements.id
membership_id	uuid / NO	DB-023 org_memberships.id

Maydon	Type / NULL	PK yoki FK
assigned_by_id	uuid / NO	DB-001 users.id
assigned_at	timestamptz / NO	-
accepted_at	timestamptz / YES	-
revoked_at	timestamptz / YES	-

INDEX: placement_id,revoked_at; membership_id,revoked_at

Faqat host orgidagi faol MENTOR. Self-review yo‘q; bir vaqtda bitta asosiy accepted mentor partial unique bilan.

Imkoniyatlar: CAP-017, CAP-018, CAP-023.

DB-052 academic_assignments

O‘qituvchining ayni amaliyotga qabuli.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
placement_id	uuid / NO	DB-049 placements.id
teacher_membership_id	uuid / NO	DB-030 institution_memberships.id
supervision_id	uuid / NO	DB-034 student_supervisions.id
assigned_by_id	uuid / NO	DB-001 users.id
assigned_at	timestamptz / NO	-
accepted_at	timestamptz / YES	-
revoked_at	timestamptz / YES	-

INDEX: placement_id,revoked_at; teacher_membership_id,revoked_at

Teacher muassasasi enrollment va supervisionga mos. Bir vaqtda bitta asosiy accepted teacher. O‘z kundaligini boshqa rol bilan ko‘rish/tasdiqlash rad.

Imkoniyatlar: CAP-018, CAP-020, CAP-021, CAP-024.

DB-053 plan_tasks

Rejaga bog‘langan natija va topshiriq.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-

Maydon	Type / NULL	PK yoki FK
terms_id	uuid / NO	DB-050 placement_terms.id
code	text / NO	-
title	text / NO	-
expected_result	text / NO	-
required	boolean / NO	-
target_minutes	integer / YES	-

UNIQUE: terms_id,code

CHECK: target_minutes IS NULL OR target_minutes >= 0

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Imkoniyatlar: CAP-017, CAP-025.

DB-054 placement_decisions

Faollashtirish/yakun/bekor qilish dalili.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
placement_id	uuid / NO	DB-049 placements.id
actor_id	uuid / NO	DB-001 users.id
side	text / NO	-
decision	text / NO	-
terms_id	uuid / NO	DB-050 placement_terms.id
reason	text / YES	-
decided_at	timestamptz / NO	-

INDEX: placement_id,decided_at

Tegishli tomon vakolati va reja versiyasi tekshiriladi; completion shartlari yopilmaguncha COMPLETED emas.

Imkoniyatlar: CAP-017, CAP-018, CAP-024, CAP-025.

LOG - Kundalik va dalillar

DB-055 daily_logs

Placement va mahalliy sana uchun bitta kun.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
placement_id	uuid / NO	DB-049 placements.id
local_date	date / NO	-
current_revision_id	uuid / YES	DB-056 log_revisions.id
version	integer / NO	-

UNIQUE: placement_id,local_date; id,placement_id

CHECK: version > 0

INDEX: placement_id,local_date

Imkoniyatlar: CAP-021, CAP-022, CAP-023, CAP-025, CAP-026.

DB-056 log_revisions

Qoralama va topshirilgan kun versiyasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
daily_log_id	uuid / NO	DB-055 daily_logs.id
revision	integer / NO	-
terms_id	uuid / NO	DB-050 placement_terms.id
author_id	uuid / NO	DB-001 users.id
day_type	text / NO	-
summary	text / YES	-
total_minutes	integer / NO	-
status	text / NO	-
submitted_at	timestamptz / YES	-
idempotency_key	text / NO	-

UNIQUE: daily_log_id,revision; idempotency_key; id,daily_log_id

CHECK: revision > 0; total_minutes >= 0

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

total_minutes faqat task_entries SUMidan serverda; SUBMITTED immutable. Sana, placement, kechikish, overlap, reja chegaralari topshirish tranzaksiyasida.

Imkoniyatlar: CAP-021, CAP-022, CAP-023, CAP-024, CAP-026.

DB-057 log_task_entries

Kun ichidagi bajarilgan ishlar.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
revision_id	uuid / NO	DB-056 log_revisions.id
plan_task_id	uuid / YES	DB-053 plan_tasks.id
result_text	text / NO	-
minutes	integer / NO	-
start_minute	integer / YES	-
end_minute	integer / YES	-

CHECK: minutes > 0; (start_minute IS NULL AND end_minute IS NULL) OR (start_minute >= 0 AND end_minute <= 1440 AND end_minute > start_minute AND minutes = end_minute - start_minute)

INDEX: revision_id

plan_task ayni placement rejasidan. Oraliq berilgan ishlar kesishmaydi; daqiqa-only yozuvga soxta aniq overlap da’vosi qilinmaydi.

Imkoniyatlar: CAP-022, CAP-023.

DB-058 log_evidence

Topshiriq dalili uchun private fayl.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
entry_id	uuid / NO	DB-057 log_task_entries.id
file_id	uuid / NO	DB-018 files.id
visibility	text / NO	-

UNIQUE: entry_id,file_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Muallif yuklash huquqi, clean scan va host confidentiality; ko‘ruvchi assignmenti tekshiriladi.

Imkoniyatlar: CAP-022, CAP-023.

DB-059 workplace_log_reviews

Mentorning aynan revisionga qarori.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
revision_id	uuid / NO	DB-056 log_revisions.id
mentor_assignment_id	uuid / NO	DB-051 mentor_assignments.id
decision	text / NO	-
comment	text / YES	-
decided_at	timestamptz / NO	-

UNIQUE: revision_id,mentor_assignment_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

APPROVED/RETURNED; faqat qabul qilingan faol assignment. Muallifga o‘zini tasdiqlash taqiqlangan; oldingi qaror yangi revisionni tasdiqlamaydi.

Imkoniyatlar: CAP-023, CAP-026.

DB-060 academic_log_reviews

O‘qituvchining alohida fikri.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
revision_id	uuid / NO	DB-056 log_revisions.id
academic_assignment_id	uuid / NO	DB-052 academic_assignments.id
decision	text / NO	-
comment	text / YES	-
decided_at	timestamptz / NO	-

UNIQUE: revision_id,academic_assignment_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

REVIEWED/NEEDS_REVISION, workplace approval o‘rnini bosa olmaydi.

Imkoniyatlar: CAP-024, CAP-026.

DB-061 log_amendment_requests

Yopilgan/kech kundalikni o‘zgartirish.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
daily_log_id	uuid / NO	DB-055 daily_logs.id
requester_id	uuid / NO	DB-001 users.id
reason	text / NO	-
status	text / NO	-
reviewer_id	uuid / YES	DB-001 users.id
resolved_at	timestamptz / YES	-

INDEX: status,created_at

Mustaqil review va audit; texnik uzilish sababli avtomatik jazo yo‘q.

Imkoniyatlar: CAP-022.

RAT - Baholash va apellyatsiya

DB-062 rubric_versions

Versiyali baholash mezonlari.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
family	text / NO	-
version	integer / NO	-
status	text / NO	-
scale_min	integer / NO	-

Maydon	Type / NULL	PK yoki FK
scale_max	integer / NO	-
published_at	timestamptz / YES	-

UNIQUE: family,version

CHECK: version > 0; scale_max > scale_min

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Workplace/academic/employer/mentor oilalari ajratiladi. Formula o‘zgarsa yangi immutable version.

Imkoniyatlar: CAP-027, CAP-040.

DB-063 rubric_criteria

Mezon va vazn.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
rubric_id	uuid / NO	DB-062 rubric_versions.id
code	text / NO	-
title	text / NO	-
weight	integer / NO	-
required	boolean / NO	-

UNIQUE: rubric_id,code

CHECK: weight > 0 AND weight <= 100

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

ACTIVE qilishda required mezonlar yig‘indisi 100 va to‘liqlik tekshiriladi.

Imkoniyatlar: CAP-027.

DB-064 rating_submissions

Haqiqiy yakuniy munosabatdan baho.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
placement_id	uuid / YES	DB-049 placements.id

Maydon	Type / NULL	PK yoki FK
engagement_id	uuid / YES	DB-047 job_engagements.id
rubric_id	uuid / NO	DB-062 rubric_versions.id
rater_id	uuid / NO	DB-001 users.id
target_user_id	uuid / YES	DB-001 users.id
target_org_id	uuid / YES	DB-021 organizations.id
version	integer / NO	-
status	text / NO	-
comment	text / YES	-
submitted_at	timestamptz / YES	-
publish_after	timestamptz / YES	-

CHECK: num_nonnulls(placement_id,engagement_id) = 1; num_nonnulls(target_user_id,target_org_id) = 1; target_user_id IS NULL OR target_user_id <> rater_id; version > 0

INDEX: target_user_id,status; target_org_id,status

Bir rater/relation/rubric uchun bitta final; partial unique indexes. Haqiqiy completed ishtirok, target va rubrika doirasi service tekshiradi. Rating feature flag boshlang‘ich OFF.

Imkoniyatlar: CAP-027, CAP-028.

DB-065 rating_revisions

Bahoning o‘zgarmas matn va formula nusxasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
submission_id	uuid / NO	DB-064 rating_submissions.id
version	integer / NO	-
comment	text / YES	-
submitted_at	timestamptz / NO	-
rubric_snapshot	jsonb / NO	-

UNIQUE: submission_id,version

CHECK: version > 0

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Submitted revision update qilinmaydi; tuzatish yangi revision, eski ball va e’tiroz dalili saqlanadi.

Imkoniyatlar: CAP-027, CAP-028.

DB-066 rating_scores

Har mezonning dalilli bali.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
rating_revision_id	uuid / NO	DB-065 rating_revisions.id
criterion_id	uuid / NO	DB-063 rubric_criteria.id
score	integer / NO	-
evidence_revision_id	uuid / YES	DB-056 log_revisions.id

UNIQUE: rating_revision_id,criterion_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Criterion submissionning ayni rubricidan; score min/max ichida. Akademik va workplace bir songa aralashmaydi.

Imkoniyatlar: CAP-027.

DB-067 rating_appeals

Reyting nizosi va mustaqil qaror.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
rating_revision_id	uuid / NO	DB-065 rating_revisions.id
appellant_id	uuid / NO	DB-001 users.id
reason	text / NO	-
status	text / NO	-
reviewer_id	uuid / YES	DB-001 users.id
decision	text / YES	-
resolved_at	timestamptz / YES	-

INDEX: status,created_at

14kun murojaat, 5ishkun nishon; nizoli baho vaqtincha aggregate tashqarisida. Reviewer munosabat ishtirokchisi emas.

Imkoniyatlar: CAP-028.

DB-068 rating_aggregates

Qayta hisoblanadigan ommaviy kesh.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
target_user_id	uuid / YES	DB-001 users.id
target_org_id	uuid / YES	DB-021 organizations.id
rubric_id	uuid / NO	DB-062 rubric_versions.id
sample_size	integer / NO	-
score	numeric(18,2) / YES	-
window_start	date / NO	-
window_end	date / NO	-
computed_at	timestamptz / NO	-

CHECK: num_nonnulls(target_user_id,target_org_id) = 1; sample_size >= 0; window_end >= window_start

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Haqiqat manbai submission, bu kesh. n<5da score public emas; unrated 0 emas. Bir relation bir ovoz; appeals bilan qayta hisob. Partial unique target/rubric/window.

Imkoniyatlar: CAP-028, CAP-040.

BIL - Tarif, hisob va to‘lov

DB-069 products

Sotiladigan xizmat turi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
code	text / NO	-
name	text / NO	-
service_kind	text / NO	-

Maydon	Type / NULL	PK yoki FK
active	boolean / NO	-

UNIQUE: code

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Candidate search free. POST_30D va ORG_PLAN_30D pilot narxlari; amaliyot/akademik xizmat shartnoma bo‘lmasa pullik va’da qilinmaydi.

Imkoniyatlar: CAP-029, CAP-039.

DB-070 price_versions

Tarif narxining immutable nusxasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
product_id	uuid / NO	DB-069 products.id
version	integer / NO	-
amount	numeric(18,2) / NO	-
currency	text / NO	-
duration_days	integer / NO	-
tax_included	boolean / NO	-
valid_from	timestamptz / NO	-
valid_to	timestamptz / YES	-

UNIQUE: product_id,version

CHECK: amount >= 0; duration_days > 0; valid_to IS NULL OR valid_to > valid_from

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Imkoniyatlar: CAP-029, CAP-039.

DB-071 orders

Tashkilot buyurtmasi va narx snapshot.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-

Maydon	Type / NULL	PK yoki FK
organization_id	uuid / NO	DB-021 organizations.id
purchaser_id	uuid / NO	DB-001 users.id
price_id	uuid / NO	DB-070 price_versions.id
opportunity_id	uuid / YES	DB-035 opportunities.id
amount	numeric(18,2) / NO	-
currency	text / NO	-
price_snapshot	jsonb / NO	-
status	text / NO	-
idempotency_key	text / NO	-

UNIQUE: idempotency_key

CHECK: amount >= 0

INDEX: organization_id,created_at

Org billing vakolati; order e’loni ayni orgga. Stavka keyin o‘zgarsa eski order o‘zgarmaydi.

Imkoniyatlar: CAP-029, CAP-030, CAP-038.

DB-072 invoices

Hisob va fiskal hujjat holati.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
order_id	uuid / NO	DB-071 orders.id
number	text / NO	-
amount	numeric(18,2) / NO	-
currency	text / NO	-
fiscal_status	text / NO	-
fiscal_reference	text / YES	-
issued_at	timestamptz / NO	-
paid_at	timestamptz / YES	-

UNIQUE: number; order_id

CHECK: amount >= 0

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Fiskal talab operatorning huquqiy shakliga mos mutaxassis bilan; oddiy PDFni fiskal chek deb atamaslik.

Imkoniyatlar: CAP-029, CAP-038.

DB-073 payment_attempts

Provayder tranzaksiyasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
order_id	uuid / NO	DB-071 orders.id
provider	text / NO	-
provider_transaction_id	text / YES	-
amount	numeric(18,2) / NO	-
currency	text / NO	-
status	text / NO	-
requested_at	timestamptz / NO	-
confirmed_at	timestamptz / YES	-

UNIQUE: provider,provider_transaction_id

CHECK: amount >= 0

INDEX: order_id,status

Bir order ko‘p attempt; muvaffaqiyat entitlementsni bir marta beradi. Callback protokoli provayderga mos; brauzer success pul tasdig‘i emas.

Imkoniyatlar: CAP-029, CAP-030.

DB-074 payment_events

Takror keluvchi callback deduplikatsiyasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
attempt_id	uuid / NO	DB-073 payment_attempts.id
provider	text / NO	-

Maydon	Type / NULL	PK yoki FK
event_key	text / NO	-
payload_digest	text / NO	-
normalized_payload	jsonb / NO	-
processed_at	timestamptz / YES	-
received_at	timestamptz / NO	-

UNIQUE: provider,event_key

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Server auth/protokol, amount/order/currency mosligi. Kartaning maxfiy rekvizitlari saqlanmaydi; payload PII minimallashtirilgan.

Imkoniyatlar: CAP-030.

DB-075 service_entitlements

Amaldagi xizmat, muddat va kvota.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
organization_id	uuid / NO	DB-021 organizations.id
order_id	uuid / YES	DB-071 orders.id
opportunity_id	uuid / YES	DB-035 opportunities.id
service_kind	text / NO	-
status	text / NO	-
starts_at	timestamptz / YES	-
ends_at	timestamptz / YES	-
quota	integer / YES	-
granted_by_id	uuid / YES	DB-001 users.id
grant_reason	text / YES	-

CHECK: ends_at IS NULL OR starts_at IS NULL OR ends_at > starts_at; quota IS NULL OR quota >= 0

INDEX: organization_id,status,ends_at

POST_30D muddati publishdan, pul tushgan kundan emas. Org plan birinchi publish yoki aniq aktivatsiyadan boshlanadi; auto-renew MVPda OFF. Bepul pilot grant auditlanadi.

Imkoniyatlar: CAP-016, CAP-029, CAP-038.

DB-076 refunds

Qaytarish va ikkita tasdiq.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
payment_attempt_id	uuid / NO	DB-073 payment_attempts.id
requester_id	uuid / NO	DB-001 users.id
approver_id	uuid / YES	DB-001 users.id
amount	numeric(18,2) / NO	-
currency	text / NO	-
reason	text / NO	-
status	text / NO	-
provider_refund_id	text / YES	-
resolved_at	timestamptz / YES	-

UNIQUE: provider_refund_id

CHECK: amount > 0; approver_id IS NULL OR approver_id <> requester_id

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Yig‘indi muvaffaqiyatli refunds captured summadan oshmaydi; lock + callback idempotency. Katta summaga ikkinchi mustaqil tasdiq audit_eventda.

Imkoniyatlar: CAP-030.

DB-077 billing_ledger

Pul va kreditning append-only jurnali.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
organization_id	uuid / NO	DB-021 organizations.id
order_id	uuid / YES	DB-071 orders.id
payment_attempt_id	uuid / YES	DB-073 payment_attempts.id
refund_id	uuid / YES	DB-076 refunds.id

Maydon	Type / NULL	PK yoki FK
entry_kind	text / NO	-
amount	numeric(18,2) / NO	-
currency	text / NO	-
occurred_at	timestamptz / NO	-
dedupe_key	text / NO	-

UNIQUE: dedupe_key

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Capture/refund/credit manbasi ko‘rsatiladi; balans tahrirlanmaydi. Reconciliation bank/provider natijasiga; bu nomzod ish haqi yoki escrow hamyoni emas.

Imkoniyatlar: CAP-029, CAP-030, CAP-038.

OPS - Moderatsiya, yordam va maxfiylik

DB-078 moderation_cases

E’lon, tashkilot, fayl yoki bahoga shikoyat.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
reporter_id	uuid / NO	DB-001 users.id
opportunity_id	uuid / YES	DB-035 opportunities.id
organization_id	uuid / YES	DB-021 organizations.id
rating_id	uuid / YES	DB-064 rating_submissions.id
file_id	uuid / YES	DB-018 files.id
category	text / NO	-
status	text / NO	-
severity	text / NO	-
assigned_to_id	uuid / YES	DB-001 users.id
due_at	timestamptz / NO	-
decision	text / YES	-
decided_at	timestamptz / YES	-

CHECK: num_nonnulls(opportunity_id,organization_id,rating_id,file_id) = 1

INDEX: status,severity,due_at

Target FK bilan. Avtomatik signal navbat yaratadi; sudsiz qora ro‘yxat yoki pulga yashirish yo‘q. Scam bo‘lsa vaqtli hide va mustaqil review.

Imkoniyatlar: CAP-028, CAP-032, CAP-033.

DB-079 moderation_actions

Cheklash/ochish/apellyatsiya tarixi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
case_id	uuid / NO	DB-078 moderation_cases.id
actor_id	uuid / NO	DB-001 users.id
action	text / NO	-
reason	text / NO	-
expires_at	timestamptz / YES	-
appealed_action_id	uuid / YES	DB-079 moderation_actions.id
occurred_at	timestamptz / NO	-

INDEX: case_id,occurred_at

Append-only; apellyatsiyani dastlabki qaror egasi yakka hal qilmaydi. E’lon yashirilsa accepted munosabat/o‘z dalili yo‘qolmaydi.

Imkoniyatlar: CAP-032, CAP-033.

DB-080 support_tickets

Foydalanuvchi yordami.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
requester_id	uuid / NO	DB-001 users.id
category	text / NO	-
priority	text / NO	-
status	text / NO	-

Maydon	Type / NULL	PK yoki FK
assignee_id	uuid / YES	DB-001 users.id
subject	text / NO	-
first_response_due_at	timestamptz / NO	-
resolved_at	timestamptz / YES	-

INDEX: status,priority,first_response_due_at; requester_id,created_at

Ruxsatli yordam xodimi zarur qismni ko‘radi; parol/OTP so‘ramaydi.

Imkoniyatlar: CAP-034.

DB-081 support_messages

Yordam yozishmasi va ilova.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
ticket_id	uuid / NO	DB-080 support_tickets.id
author_id	uuid / NO	DB-001 users.id
body	text / NO	-
file_id	uuid / YES	DB-018 files.id
internal	boolean / NO	-

INDEX: ticket_id,created_at

Internal note foydalanuvchi eksportiga chiqmaydi; ilova ayni ticketga ruxsat bilan.

Imkoniyatlar: CAP-034.

DB-082 privacy_requests

Eksport, tuzatish va o‘chirish.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
user_id	uuid / NO	DB-001 users.id
kind	text / NO	-

Maydon	Type / NULL	PK yoki FK
status	text / NO	-
verified_at	timestamptz / YES	-
due_at	timestamptz / NO	-
result_file_id	uuid / YES	DB-018 files.id
resolved_at	timestamptz / YES	-
reason	text / YES	-

INDEX: status,due_at

Eksport qayta auth, qisqa TTL link. O‘chirish moliya/legal hold bilan retention siyosatiga muvofiq; muddat huquqiy tasdiqdan so‘ng.

Imkoniyatlar: CAP-035.

DB-083 retention_policies

Ma’lumot toifasi va saqlash rejasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
data_class	text / NO	-
policy_version	integer / NO	-
active_days	integer / YES	-
backup_expiry_days	integer / NO	-
legal_basis	text / NO	-
status	text / NO	-

UNIQUE: data_class,policy_version

CHECK: active_days IS NULL OR active_days >= 0; backup_expiry_days > 0

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Taklif va qonuniy muddat ajratiladi; NULL muddatsiz saqlash ruxsati emas, qaror kutilmoqda. Legal approval bo‘lmagan toifa pilotda yig‘ilmaydi.

Imkoniyatlar: CAP-035.

DB-084 legal_holds

Nizo uchun minimal saqlash to‘xtatgichi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
user_id	uuid / YES	DB-001 users.id
organization_id	uuid / YES	DB-021 organizations.id
case_id	uuid / YES	DB-078 moderation_cases.id
reason	text / NO	-
authorized_by_id	uuid / NO	DB-001 users.id
expires_at	timestamptz / NO	-
released_at	timestamptz / YES	-

CHECK: num_nonnulls(user_id,organization_id,case_id) = 1

INDEX: expires_at,released_at

Vakolatli asos va aniq muddat; barcha hisobni cheksiz saqlash uchun bahona emas.

Imkoniyatlar: CAP-035.

SYS - Xabar, audit va boshqaruv

DB-085 notification_preferences

Kanal va maqsad sozlamasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
user_id	uuid / NO	DB-001 users.id
channel	text / NO	-
purpose	text / NO	-
enabled	boolean / NO	-
consent_event_id	uuid / YES	DB-011 consent_events.id

UNIQUE: user_id,channel,purpose

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Marketing opt-in; zarur statuslar sayt ichida qoladi. External kanal tanloviga hurmat qilinadi.

Imkoniyatlar: CAP-031.

DB-086 outbox_events

DB tranzaksiyasi bilan chiqadigan hodisa.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
event_type	text / NO	-
schema_version	integer / NO	-
aggregate_key	text / NO	-
payload	jsonb / NO	-
dedupe_key	text / NO	-
available_at	timestamptz / NO	-
processed_at	timestamptz / YES	-
attempts	integer / NO	-

UNIQUE: dedupe_key

CHECK: schema_version > 0; attempts >= 0

INDEX: processed_at,available_at

Append-only biznes hodisa; payload PII emas yoki minimal opaque ID. Consumer eng kamida bir marta kelishiga chidamli; aggregate_key audit/routing uchun, domain FK o‘rniga emas.

Imkoniyatlar: CAP-031, CAP-037.

DB-087 notifications

Foydalanuvchiga sayt xabari.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
user_id	uuid / NO	DB-001 users.id
event_id	uuid / NO	DB-086 outbox_events.id
template_code	text / NO	-

Maydon	Type / NULL	PK yoki FK
opportunity_id	uuid / YES	DB-035 opportunities.id
application_id	uuid / YES	DB-041 applications.id
placement_id	uuid / YES	DB-049 placements.id
ticket_id	uuid / YES	DB-080 support_tickets.id
read_at	timestamptz / YES	-

UNIQUE: user_id,event_id,template_code

INDEX: user_id,read_at,created_at

Havola ochilganda joriy huquq qayta tekshiriladi; xabarda maxfiy CV/log matni yo‘q.

Imkoniyatlar: CAP-031.

DB-088 notification_deliveries

Kanalga yuborish urinishlari.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
notification_id	uuid / NO	DB-087 notifications.id
channel	text / NO	-
provider	text / NO	-
provider_message_id	text / YES	-
status	text / NO	-
attempt_no	integer / NO	-
next_retry_at	timestamptz / YES	-
delivered_at	timestamptz / YES	-
error_code	text / YES	-

UNIQUE: notification_id,channel,attempt_no

CHECK: attempt_no > 0

INDEX: status,next_retry_at

Provider timeout retry backoff+jitter va xarajat limiti; callback foydalanuvchi tasdiqi emas. SMS/WhatsApp delivery OTP consume qilmaydi.

Imkoniyatlar: CAP-031.

DB-089 audit_events

Vakolatli amallarning append-only izi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
actor_id	uuid / YES	DB-001 users.id
session_id	uuid / YES	DB-004 sessions.id
action	text / NO	-
object_type	text / NO	-
object_key	text / NO	-
scope_snapshot	jsonb / NO	-
before_digest	text / YES	-
after_digest	text / YES	-
reason	text / YES	-
request_id	text / NO	-
occurred_at	timestamptz / NO	-

INDEX: actor_id,occurred_at; object_type,object_key,occurred_at; request_id

Object nomi tarixiy audit identifikatori, aktiv domain FK emas. Sir/OTP/raw CV yo‘q; DB roli update/delete qilolmaydi, tashqi immutable nusxa. Aktyor pseudonymi tarixni saqlaydi.

Imkoniyatlar: CAP-026, CAP-036.

DB-090 idempotency_keys

Sezgir HTTP amallarining bir marta natijasi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
user_id	uuid / NO	DB-001 users.id
operation	text / NO	-
key	text / NO	-
request_digest	text / NO	-

Maydon	Type / NULL	PK yoki FK
response_status	integer / YES	-
response_ref	text / YES	-
expires_at	timestamptz / NO	-

UNIQUE: user_id,operation,key

INDEX: expires_at

Body digest boshqa bo‘lsa409. Token ownership, operation va scope bilan. Biznes UNIQUE constraintlar TTLdan keyin ham takror moliya yaratishni to‘sadi.

Imkoniyatlar: CAP-007, CAP-011.

DB-091 analytics_events

O‘lchov va segment, PII emas.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
event_key	text / NO	-
event_name	text / NO	-
actor_pseudonym	text / YES	-
cohort	text / YES	-
properties	jsonb / NO	-
occurred_at	timestamptz / NO	-
schema_version	integer / NO	-

UNIQUE: event_key

INDEX: event_name,occurred_at

Kontakt/log/CV yo‘q; rozilik asosli statistika. Ish boshlash faqat 2tomon tasdiq hodisasi, raw click bilan teng emas.

Imkoniyatlar: CAP-037.

DB-092 feature_flags

Xizmatni boshqariladigan ishga tushirish.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-

Maydon	Type / NULL	PK yoki FK
updated_at	timestamptz / NO	-
code	text / NO	-
enabled	boolean / NO	-
rule_schema	jsonb / NO	-
changed_by_id	uuid / NO	DB-001 users.id
reason	text / NO	-

UNIQUE: code

INDEX: PK/UNIQUE va FK leading-column indekslar; schema_loyihasi.sqlga qarang.

Default OFF: ratings, academic_official_export, WhatsApp, paid_checkout. Flag ruxsatni chetlab o‘tmaydi; o‘zgarish auditlanadi.

Imkoniyatlar: CAP-037, CAP-039, CAP-040.

DB-093 incidents

Uzilish, egalik va tiklash dalili.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
severity	text / NO	-
status	text / NO	-
commander_id	uuid / NO	DB-001 users.id
summary	text / NO	-
started_at	timestamptz / NO	-
resolved_at	timestamptz / YES	-
postmortem_ref	text / YES	-

INDEX: status,severity,started_at

24/7 yordam va’dasi faqat amaldagi navbatchi bilan; hozir P1 on-call tashkil etilishi reliz sharti.

Imkoniyatlar: CAP-036, CAP-037.

DB-094 restore_runs

Backupni tiklash tekshiruvi.

Maydon	Type / NULL	PK yoki FK
id	uuid / NO	PK
created_at	timestamptz / NO	-
updated_at	timestamptz / NO	-
started_at	timestamptz / NO	-
finished_at	timestamptz / YES	-
backup_ref	text / NO	-
status	text / NO	-
recovered_through	timestamptz / YES	-
measured_rpo_minutes	integer / YES	-
measured_rto_minutes	integer / YES	-
checked_by_id	uuid / NO	DB-001 users.id
evidence_ref	text / YES	-

INDEX: started_at

Backup fayli borligi restore testi emas. Billing yoqilishidan oldin PITR/ledger reconciliation dalili.

Imkoniyatlar: CAP-037.

API amallari kontrakt jadvali

Method + path + operation role + CAP scope birga ishlaydi. GUEST qatnashgan AUTH/INVITE oilasida POSTning o‘z auth/purpose tekshiruvi bor; taklif GET minimal, accept verified login bilan. /operator amallari faqat ko‘rsatilgan operator rolida. Bu OpenAPI JSON schema tayyorligi yoki ishlayotgan endpoint da’vosi emas.

API va method	Path	CAP / amal rollari
API-001 GET	/api/v1/opportunities	CAP-001 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-002 GET	/api/v1/onboarding/options	CAP-002 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-003 POST	/api/v1/matching/preview	CAP-005 / CANDIDATE, STUDENT
API-004 POST	/api/v1/saved-opportunities	CAP-006 / CANDIDATE, STUDENT
API-005 GET	/api/v1/opportunities/{id}	CAP-001 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-006 POST	/api/v1/opportunities/{id}/applications	CAP-007 / CANDIDATE, STUDENT
API-007 POST	/api/v1/moderation-cases	CAP-032 / CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, TEACHER, COORDINATOR
API-008 GET	/api/v1/organizations/{id}/public	CAP-001 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-009 GET	/api/v1/organizations/{id}/opportunities	CAP-001 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-010 GET	/api/v1/organizations/{id}/rating-summary	CAP-040 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR
API-011 GET	/api/v1/services/public	CAP-039 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-012 GET	/api/v1/prices/public	CAP-039 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-013 GET	/api/v1/help/{document}	CAP-039 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN

API va method	Path	CAP / amal rollari
API-014 GET	/api/v1/policies/{document}	CAP-039 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-015 POST	/api/v1/auth/register	CAP-002 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-016 POST	/api/v1/auth/challenges	CAP-002 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-017 GET	/api/v1/institution-onboarding/options	CAP-019 / COORDINATOR
API-018 POST	/api/v1/auth/login	CAP-002 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-019 POST	/api/v1/auth/mfa/verify	CAP-002 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-020 POST	/api/v1/auth/challenges/{id}/consume	CAP-002 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-021 POST	/api/v1/auth/challenges/{id}/resend	CAP-002 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-022 POST	/api/v1/auth/recovery	CAP-002 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-023 GET	/api/v1/auth/recovery/{id}	CAP-002 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-024 GET	/api/v1/invitations/{token}	CAP-015 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-025 POST	/api/v1/invitations/{token}/accept	CAP-015 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-026 PATCH	/api/v1/me/candidate	CAP-003 / CANDIDATE, STUDENT

API va method	Path	CAP / amal rollari
API-027 POST	/api/v1/me/enrollments	CAP-004 / CANDIDATE, STUDENT
API-028 PUT	/api/v1/me/availability	CAP-005 / CANDIDATE, STUDENT
API-029 GET	/api/v1/me/applications/summary	CAP-007 / CANDIDATE, STUDENT
API-030 GET	/api/v1/me/placements	CAP-018 / CANDIDATE, STUDENT, MENTOR, TEACHER
API-031 GET	/api/v1/me/notifications	CAP-031 / CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-032 PATCH	/api/v1/me/candidate/skills	CAP-003 / CANDIDATE, STUDENT
API-033 POST	/api/v1/me/candidate/cv	CAP-003 / CANDIDATE, STUDENT
API-034 GET	/api/v1/me/enrollments	CAP-004 / CANDIDATE, STUDENT
API-035 GET	/api/v1/me/availability	CAP-005 / CANDIDATE, STUDENT
API-036 PUT	/api/v1/me/availability/exceptions	CAP-005 / CANDIDATE, STUDENT
API-037 GET	/api/v1/me/saved-opportunities	CAP-006 / CANDIDATE, STUDENT
API-038 DELETE	/api/v1/me/saved-opportunities/{id}	CAP-006 / CANDIDATE, STUDENT
API-039 GET	/api/v1/me/applications	CAP-007 / CANDIDATE, STUDENT
API-040 POST	/api/v1/me/applications/{id}/reapply	CAP-007 / CANDIDATE, STUDENT
API-041 GET	/api/v1/me/applications/{id}	CAP-007 / CANDIDATE, STUDENT
API-042 POST	/api/v1/interviews/{id}/responses	CAP-009 / CANDIDATE, STUDENT, OWNER, HR
API-043 POST	/api/v1/offers/{id}/respond	CAP-011 / CANDIDATE, STUDENT
API-044 POST	/api/v1/me/applications/{id}/withdraw	CAP-007 / CANDIDATE, STUDENT
API-045 GET	/api/v1/me/engagements/{id}	CAP-012 / CANDIDATE, STUDENT, OWNER, HR
API-046 POST	/api/v1/engagements/{id}/confirmations	CAP-012 / CANDIDATE, STUDENT, OWNER, HR
API-047 POST	/api/v1/engagements/{id}/ratings	CAP-027 / CANDIDATE, STUDENT, HR, HOST_COORDINATOR, MENTOR, TEACHER
API-048 POST	/api/v1/placements/{id}/accept	CAP-018 / CANDIDATE, STUDENT, MENTOR, TEACHER
API-049 POST	/api/v1/placements/{id}/completion-requests	CAP-025 / CANDIDATE, STUDENT, HOST_COORDINATOR, MENTOR, TEACHER, COORDINATOR
API-050 GET	/api/v1/placements/{id}/plan	CAP-018 / CANDIDATE, STUDENT, MENTOR, TEACHER
API-051 POST	/api/v1/placements/{id}/assignments/{assignment}/accept	CAP-018 / CANDIDATE, STUDENT, MENTOR, TEACHER

API va method	Path	CAP / amal rollari
API-052 POST	/api/v1/placements/{id}/exports	CAP-026 / CANDIDATE, STUDENT, MENTOR, TEACHER, COORDINATOR
API-053 GET	/api/v1/placements/{id}/logs	CAP-022 / CANDIDATE, STUDENT
API-054 PUT	/api/v1/placements/{id}/logs/{date}	CAP-022 / CANDIDATE, STUDENT
API-055 POST	/api/v1/log-entries/{id}/evidence	CAP-022 / CANDIDATE, STUDENT
API-056 POST	/api/v1/logs/{id}/submit	CAP-022 / CANDIDATE, STUDENT
API-057 GET	/api/v1/me/ratings	CAP-027 / CANDIDATE, STUDENT, HR, HOST_COORDINATOR, MENTOR, TEACHER
API-058 POST	/api/v1/ratings/{id}/appeals	CAP-028 / CANDIDATE, STUDENT, OWNER, HR, MENTOR, TEACHER, MODERATOR
API-059 PUT	/api/v1/me/notification-preferences	CAP-031 / CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-060 PATCH	/api/v1/me/security	CAP-002 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-061 DELETE	/api/v1/me/sessions/{id}	CAP-002 / GUEST, CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-062 POST	/api/v1/me/privacy-requests	CAP-035 / CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-063 GET	/api/v1/me/support-tickets	CAP-034 / CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-064 POST	/api/v1/support-tickets/{id}/messages	CAP-034 / CANDIDATE, STUDENT, OWNER, HR, HOST_COORDINATOR, MENTOR, BILLING, COORDINATOR, TEACHER, MODERATOR, SUPPORT, FINANCE, ADMIN
API-065 POST	/api/v1/organizations	CAP-013 / OWNER, HOST_COORDINATOR
API-066 POST	/api/v1/organizations/{id}/verification	CAP-013 / OWNER, HOST_COORDINATOR
API-067 GET	/api/v1/organizations/{id}/verification	CAP-013 / OWNER, HOST_COORDINATOR
API-068 GET	/api/v1/organizations/{org}/dashboard	CAP-008 / OWNER, HR, HOST_COORDINATOR
API-069 GET	/api/v1/organizations/{org}/placements	CAP-017 / OWNER, HOST_COORDINATOR
API-070 GET	/api/v1/organizations/{org}/entitlements	CAP-029 / OWNER, BILLING

API va method	Path	CAP / amal rollari
API-071 PATCH	/api/v1/organizations/{org}	CAP-013 / OWNER, HOST_COORDINATOR
API-072 GET	/api/v1/organizations/{org}/capabilities	CAP-013 / OWNER, HOST_COORDINATOR
API-073 GET	/api/v1/organizations/{org}/memberships	CAP-014 / OWNER
API-074 POST	/api/v1/organizations/{org}/invitations	CAP-014 / OWNER
API-075 POST	/api/v1/organizations/{org}/memberships/{id}/revoke	CAP-014 / OWNER
API-076 GET	/api/v1/organizations/{org}/opportunities	CAP-016 / OWNER, HR, HOST_COORDINATOR
API-077 POST	/api/v1/opportunities/{id}/transition	CAP-016 / OWNER, HR, HOST_COORDINATOR
API-078 PUT	/api/v1/opportunities/{id}/draft	CAP-016 / OWNER, HR, HOST_COORDINATOR
API-079 PUT	/api/v1/opportunities/{id}/draft/terms	CAP-016 / OWNER, HR, HOST_COORDINATOR
API-080 PUT	/api/v1/opportunities/{id}/draft/schedule	CAP-016 / OWNER, HR, HOST_COORDINATOR
API-081 POST	/api/v1/opportunities/{id}/publish	CAP-016 / OWNER, HR, HOST_COORDINATOR
API-082 GET	/api/v1/opportunities/{id}/applications	CAP-008 / OWNER, HR, HOST_COORDINATOR
API-083 POST	/api/v1/applications/{id}/transition	CAP-008 / OWNER, HR, HOST_COORDINATOR
API-084 GET	/api/v1/applications/{id}/hr	CAP-008 / OWNER, HR, HOST_COORDINATOR
API-085 POST	/api/v1/applications/{id}/interviews	CAP-009 / CANDIDATE, STUDENT, OWNER, HR
API-086 POST	/api/v1/applications/{id}/offers	CAP-010 / OWNER, HR, HOST_COORDINATOR
API-087 POST	/api/v1/placements/{id}/mentor-assignments	CAP-017 / OWNER, HOST_COORDINATOR
API-088 POST	/api/v1/placements/{id}/host-decisions	CAP-025 / CANDIDATE, STUDENT, HOST_COORDINATOR, MENTOR, TEACHER, COORDINATOR
API-089 GET	/api/v1/me/mentor-assignments	CAP-018 / CANDIDATE, STUDENT, MENTOR, TEACHER
API-090 GET	/api/v1/me/mentor-review-queue	CAP-023 / MENTOR
API-091 GET	/api/v1/log-revisions/{revision}/workplace	CAP-023 / MENTOR
API-092 POST	/api/v1/log-revisions/{revision}/workplace-reviews	CAP-023 / MENTOR
API-093 GET	/api/v1/organizations/{org}/billing	CAP-029 / OWNER, BILLING
API-094 POST	/api/v1/organizations/{org}/orders	CAP-029 / OWNER, BILLING
API-095 POST	/api/v1/orders/{id}/refund-requests	CAP-030 / OWNER, BILLING, FINANCE
API-096 POST	/api/v1/institution-appointments	CAP-019 / COORDINATOR
API-097 GET	/api/v1/institution-appointments/{id}	CAP-019 / COORDINATOR
API-098 GET	/api/v1/institutions/{inst}/supervisions	CAP-021 / COORDINATOR, TEACHER

API va method	Path	CAP / amal rollari
API-099 GET	/api/v1/institutions/{inst}/assignment-queue	CAP-020 / COORDINATOR
API-100 POST	/api/v1/institutions/{inst}/exports	CAP-026 / CANDIDATE, STUDENT, MENTOR, TEACHER, COORDINATOR
API-101 POST	/api/v1/student-supervisions	CAP-020 / COORDINATOR
API-102 POST	/api/v1/placements/{id}/academic-assignments	CAP-020 / COORDINATOR
API-103 POST	/api/v1/academic-assignments/{id}/revoke	CAP-020 / COORDINATOR
API-104 GET	/api/v1/me/academic-assignments	CAP-018 / CANDIDATE, STUDENT, MENTOR, TEACHER
API-105 GET	/api/v1/me/academic-review-queue	CAP-021 / COORDINATOR, TEACHER
API-106 GET	/api/v1/log-revisions/{revision}/academic	CAP-021 / COORDINATOR, TEACHER
API-107 POST	/api/v1/log-revisions/{revision}/academic-reviews	CAP-024 / TEACHER
API-108 POST	/api/v1/placements/{id}/academic-decisions	CAP-024 / TEACHER
API-109 GET	/api/v1/operator/verification-cases	CAP-033 / MODERATOR
API-110 POST	/api/v1/operator/verification-cases/{id}/decide	CAP-033 / MODERATOR
API-111 POST	/api/v1/operator/moderation-cases/{id}/actions	CAP-033 / MODERATOR
API-112 GET	/api/v1/operator/support-tickets	CAP-034 / SUPPORT
API-113 POST	/api/v1/operator/support-tickets/{id}/messages	CAP-034 / SUPPORT
API-114 POST	/api/v1/operator/privacy-requests/{id}/resolve	CAP-035 / SUPPORT
API-115 GET	/api/v1/operator/reconciliation	CAP-030 / FINANCE
API-116 POST	/api/v1/operator/refunds/{id}/approve	CAP-030 / FINANCE
API-117 GET	/api/v1/operator/finance-summary	CAP-038 / FINANCE
API-118 GET	/api/v1/operator/audit-events	CAP-036 / ADMIN
API-119 POST	/api/v1/operator/access-requests	CAP-036 / ADMIN
API-120 POST	/api/v1/operator/users/{id}/revoke-sessions	CAP-036 / ADMIN
API-121 PATCH	/api/v1/operator/feature-flags/{code}	CAP-037 / ADMIN
API-122 POST	/api/v1/operator/incidents	CAP-037 / ADMIN
API-123 GET	/api/v1/operator/restore-runs	CAP-037 / ADMIN
API-124 GET	/api/v1/operator/analytics	CAP-037 / ADMIN

Umumiy yordamchi APIlar

Quyidagi APIlar sahifa bo‘limlarida fayl/OTP/worker yordamchi amallari sifatida qo‘llanadi; alohida mahsulot sahifasi emas.

Kod	Method/path	Shart
API-X01	POST /api/v1/files/upload-intents	Owner, maqsad va scope; size/MIME allowlist
API-X02	POST /api/v1/files/{id}/finalize	Hash, byte size va karantin scan; boshqa user file rad
API-X03	GET /api/v1/files/{id}/download	Joriy scope, clean scan, short URL/proxy va audit
API-X04	POST /api/v1/auth/logout	Joriy sessiya revoke; local private draft clear
API-X05	POST /api/v1/webhooks/payments/{provider}	Provayderning aniq auth/protokoli; event dedupe/order/amount
API-X06	POST /api/v1/webhooks/delivery/{provider}	Verified provider callback; delivery login proof emas
API-X07	GET /api/v1/health/ready	Public minimal healthy/unhealthy; ichki secrets/DB tafsiloti yo‘q
API-X08	POST /api/v1/me/notifications/{id}/read	O‘z notificationi; retry idempotent